

Aromatic heterocycles 1: reactions

29

Connections

➡ Building on

- Aromaticity **ch7**
- Enols and enolates **ch20**
- Electrophilic aromatic substitution **ch21**
- Nucleophilic attack on aromatic rings **ch22**
- Reactions of enols and enolates **ch25 & ch26**

Arriving at

- Aromatic systems conceptually derived from benzene: replacing CH with N to get pyridine
- Replacing CH=CH with N to get pyrrole
- How pyridine reacts
- How pyridine derivatives can be used to extend pyridine's reactivity
- How pyrrole reacts
- How furan and thiophene compare with pyrrole
- Putting more nitrogens in five- and six-membered rings
- Fused rings: indole, quinoline, isoquinoline, and indolizine
- Rings with nitrogen and another heteroatom: oxygen or sulfur

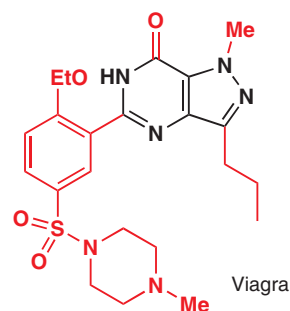
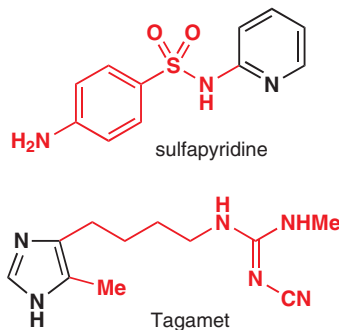
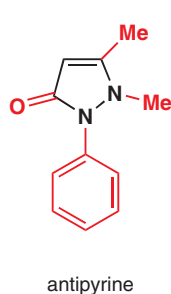
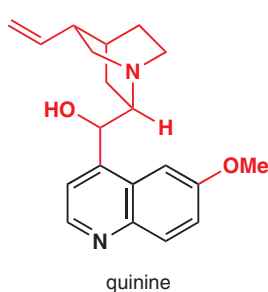
➡ Looking forward to

- Synthesis of aromatic heterocycles **ch30**
- Saturated heterocycles **ch31**
- Biological chemistry **ch42**

Introduction

Benzene is aromatic because it has six electrons in a cyclic conjugated system. We know it is aromatic because it is exceptionally stable, it has a ring current and hence large chemical shifts in the proton NMR spectrum, and it has special chemistry involving substitution rather than addition with electrophiles. This chapter and the next are about the very large number of other aromatic systems in which one or more atoms in the benzene ring are replaced by heteroatoms such as N, O, and S. There are thousands of these systems with five- and six-membered rings, and we will examine just a few.

➡ The rather precise chemical definition of 'aromatic' is explained in Chapter 7. You will find the reactions of benzene and its aromatic derivatives described in Chapters 21 and 22: those two chapters are essential reading before you tackle this one.



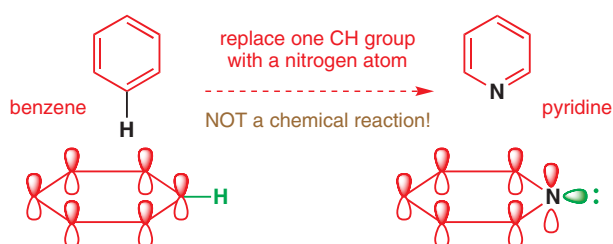
Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

Our subject is aromatic heterocycles and it is important that we treat it seriously because most—probably about two-thirds—organic compounds belong to this class, and they number among them some of the most significant compounds for human beings. If we think only of drugs we can define the history of medicine by heterocycles. Even in the sixteenth century quinine was used to prevent and treat malaria, although the structure of the drug was not known. The first synthetic drug was antipyrine (1887) for the reduction of fevers. The first effective antibiotic was sulfapyridine (1938). The first multi-million pound drug (1970s) was Tagamet, the anti-ulcer drug, and among the most topical of current drugs is Viagra (1997) for treatment of male impotence.

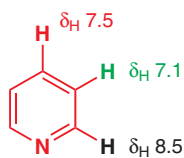
All these compounds have heterocyclic aromatic rings shown in black. Three have single rings, five- or six-membered, two have five- or six-membered rings fused together. The number of nitrogens in the rings varies from one to four. We will start by looking at the simple six-membered ring with one nitrogen atom: pyridine.

Aromaticity survives when parts of benzene's ring are replaced by nitrogen atoms

There is no doubt that benzene is aromatic. Now we must ask: how can we insert a heteroatom into the ring and retain aromaticity? What kind of atom is needed? If we want to replace one of the carbon atoms of benzene with a heteroatom, we need an atom that can be trigonal to keep the flat hexagonal ring, and that has a p orbital to keep the six delocalized electrons. Nitrogen fits all of these requirements. This is what happens if we replace a CH group in benzene with a nitrogen atom.



Interactive structure of pyridine



¹H NMR spectrum of pyridine

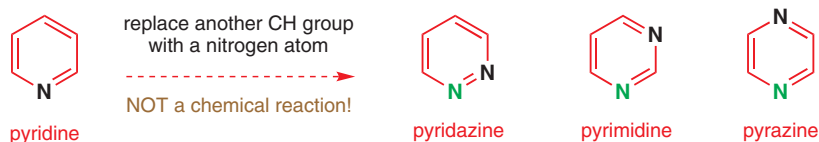
Nomenclature

One of the most annoying things about heterocyclic chemistry is the mass of what appear to be illogical names. You should not, of course, attempt to learn them all, but a basic idea of how they are designed will help you. We will give you a guide on which names to learn shortly. For the moment accept that 'amine' ends in '-ine' and any heterocyclic compound whose name ends in '-ine' is a nitrogen heterocycle. The syllable 'azo-' also implies nitrogen and 'pyr-' (usually) implies a six-membered ring (except in pyrrole!).

The orbitals in the ring have not changed in position or shape and we still have the six electrons from the three double bonds. One obvious difference is that nitrogen is trivalent and thus there is no NH bond. Instead, a lone pair of electrons occupies the space of the C–H bond in benzene.

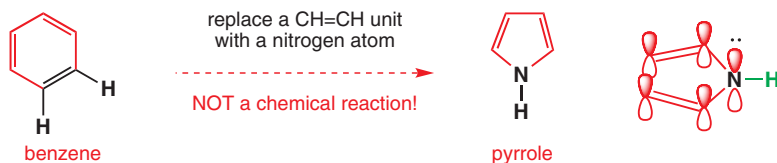
In theory then, pyridine is aromatic. But is it in real life? The most important evidence comes from the proton NMR spectrum. The six protons of benzene resonate at 7.27 ppm, some 2 ppm downfield from the alkene region, clear evidence for a ring current (Chapter 13). Pyridine is not as symmetrical as benzene but the three types of proton all resonate in the same region. As we will see, pyridine is also very stable and, by any reasonable assessment, pyridine is aromatic.

We could continue the process of replacing, on paper, more CH groups with nitrogen atoms, and would find three new aromatic heterocycles: pyridazine, pyrimidine, and pyrazine:



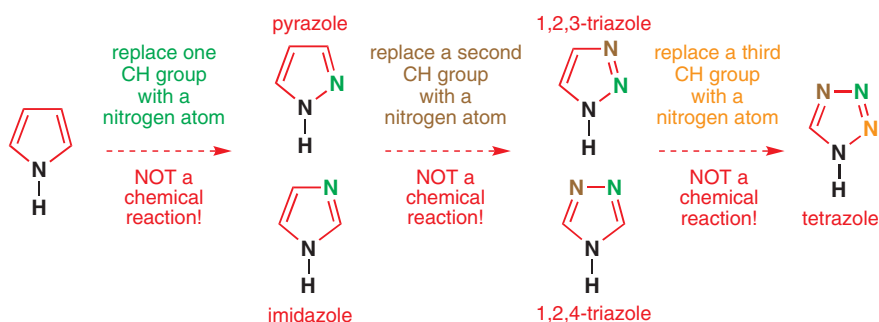
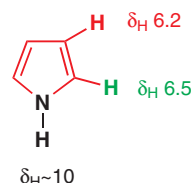
There is another way in which we might transform benzene into a heterocycle. Instead of using just one electron from N to replace an electron in the π system, we could use nitrogen's lone pair of electrons to replace two electrons in the π system. We can substitute a CH=CH unit in benzene with a nitrogen atom providing that we can use the lone pair in the delocalized system. This means putting it into a p orbital. We still have the four electrons from the

remaining double bonds and, with the two electrons of the lone pair on nitrogen, that makes six in all. The nitrogen atom must still be trigonal with the lone pair in a p orbital so the N–H bond is in the plane of the five-membered ring.



The ^1H NMR spectrum of pyrrole is slightly less convincing as the two types of proton on the ring resonate at higher field (6.5 and 6.2 ppm) than those of benzene or pyridine but they still fall in the aromatic rather than the alkene region. Pyrrole is also more reactive towards electrophiles than benzene or pyridine, but it does the usual aromatic substitution reactions (Friedel–Crafts, nitration, halogenation) rather than addition reactions: pyrrole is also aromatic.

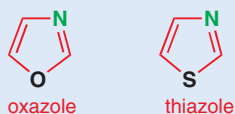
Inventing heterocycles by further replacement of CH groups by nitrogen in pyrrole leads to two compounds, pyrazole and imidazole, after one replacement, to two triazoles after two replacements, and to a single tetrazole after three.



All of these compounds are generally accepted as aromatic too as they broadly have the NMR spectra and reactivities expected for aromatic compounds. As you may expect, introducing heteroatoms into the aromatic ring and, even more, changing the ring size actually affect the chemistry a great deal. We must now return to pyridine and work our way more slowly through the chemistry of these important heterocycles to establish the principles that govern their behaviour.

More nomenclature

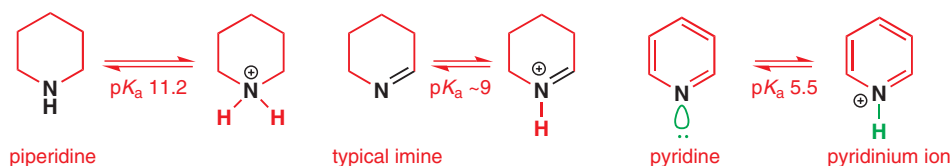
The ending ‘-ole’ is systematic and refers to a five-membered heterocyclic ring. All the five-membered aromatic heterocycles with nitrogen in the ring are sometimes called ‘the azoles’. Oxazole and thiazole are used for the oxygen and sulfur analogues of imidazole.



Pyridine is a very unreactive aromatic imine

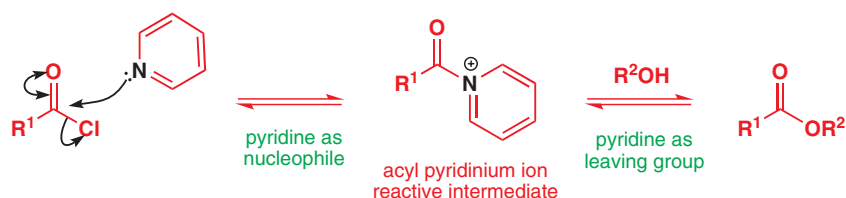
The nitrogen atom in the pyridine ring is planar and trigonal with the lone pair in the plane of the ring. This makes it an imine. Most of the imines you have met before (in Chapter 11, for example), have been unstable intermediates in carbonyl group reactions, but in pyridine we have a stable imine—stable because of its aromaticity. All imines are more weakly basic than saturated amines and pyridine is a weak base with a $\text{p}K_{\text{a}}$ (for its conjugate acid) of 5.5. This means that the pyridinium ion is about as strong an acid as a carboxylic acid.

Pyridine is also toxic and has a foul smell—so there are disadvantages in using pyridine as a solvent. But it is cheap and remains a popular solvent in spite of the problems.



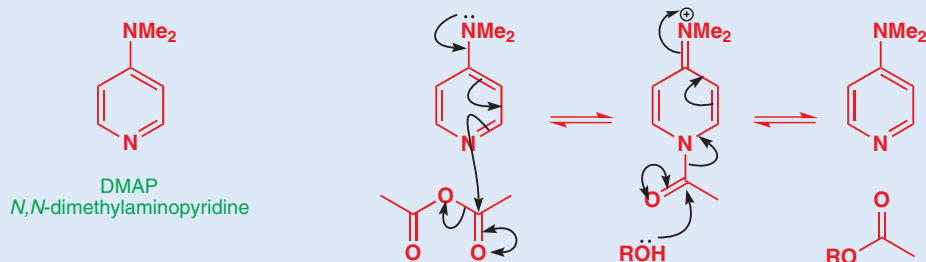
Pyridine is a reasonable nucleophile for carbonyl groups and is often used as a nucleophilic catalyst in acylation reactions. Esters are often made in pyridine solution from alcohols and acid chlorides (the full mechanism is on p. 199 of Chapter 10).

Interactive mechanism for pyridine nucleophilic catalysis

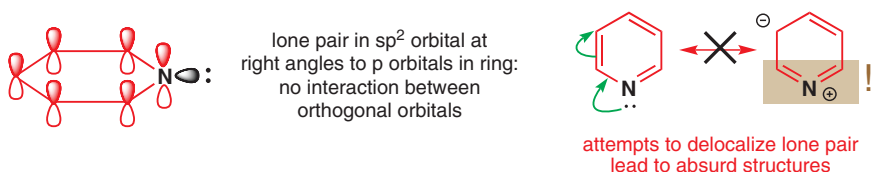


DMAP

One particular amino-pyridine has a special role as a more effective acylation catalyst than pyridine itself. This is DMAP (*N,N*-dimethylaminopyridine) in which the amino group is placed to reinforce the nucleophilic nature of the nitrogen atom. Whereas acylations 'catalysed' by pyridine are normally carried out in solution in pyridine, only small amounts of DMAP in other solvents are needed to do the same job.



Pyridine is nucleophilic at the nitrogen atom because *the lone pair of electrons on nitrogen cannot be delocalized around the ring*. They are in an sp^2 orbital orthogonal to the p orbitals in the ring and there is no interaction between orthogonal orbitals. Try it for yourself, drawing arrows. All attempts to delocalize the electrons lead to impossible results!



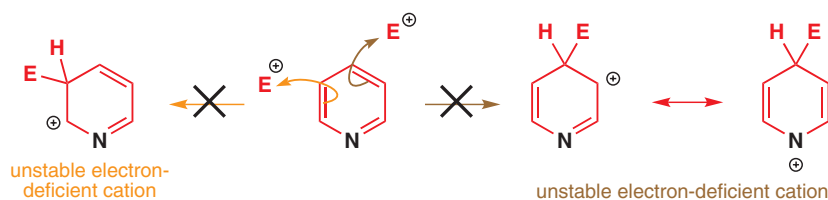
- The lone pair of pyridine's nitrogen atom is not delocalized.

Our main question about the reactivity of pyridine must be this: what does the nitrogen atom do to the rest of the ring? The important orbitals—the p orbitals of the aromatic system—are superficially the same as in benzene, but the more electronegative nitrogen atom will lower the energy of all the orbitals. Lower-energy filled orbitals mean a *less* reactive nucleophile but a lower-energy LUMO means a *more* reactive electrophile. This is a good guide to the chemistry

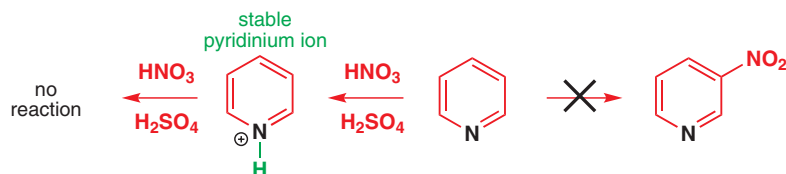
of pyridine. It is less reactive than benzene in electrophilic aromatic substitution reactions, but nucleophilic substitution, which is difficult for benzene, comes easily to pyridine.

Pyridine is bad at electrophilic aromatic substitution

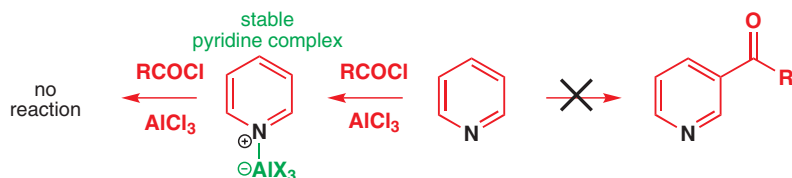
The lower energy of the orbitals of pyridine's π system means that electrophilic attack on the ring is difficult. Another way to look at this is to see that the nitrogen atom destabilizes the cationic would-be intermediate, especially when it can be delocalized onto nitrogen.



An equally serious problem is that the nitrogen lone pair is basic and a reasonably good nucleophile—this is the basis for its role as a nucleophilic catalyst in acylations. The normal reagents for electrophilic substitution reactions, such as nitration, are acidic. Treatment of pyridine with the usual mixture of HNO_3 and H_2SO_4 merely protonates the nitrogen atom. Pyridine itself is not very reactive towards electrophiles: the pyridinium ion is totally unreactive.



Other reactions, such as Friedel–Crafts acylations, require Lewis acids and these too react at nitrogen. Pyridine is a good ligand for metals such as Al(III) or Sn(IV) and, once again, the complex with its cationic nitrogen is completely unreactive towards electrophiles.



● Pyridine does not undergo electrophilic substitution

Aromatic electrophilic substitution on pyridine is not a useful reaction. The ring is unreactive and the electrophilic reagents attack nitrogen, making the ring even less reactive. Avoid nitration, sulfonation, halogenation, and Friedel–Crafts reactions on simple pyridines.

Nucleophilic substitution is easy with pyridines

By contrast, the nitrogen atom makes pyridines *more* reactive towards nucleophilic substitution, particularly at the 2- and 4-positions, by lowering the LUMO energy of the π system of pyridine. You can see this effect in action in the ease of replacement of halogens in these positions by nucleophiles.

➡ Electrophilic substitution in benzene is discussed in Chapter 21.

■ Contrast the unstable electron-deficient cationic intermediate with the stable pyridinium ion. The nitrogen lone pair is used to make the pyridinium ion but is not involved in the unstable intermediate. Note that reaction at the 3-position is the best option but still doesn't occur. Reaction at the 2- and 4-positions is worse.

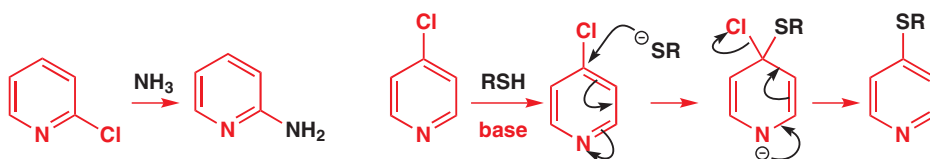
➡ Nucleophilic substitution in benzene is discussed in Chapter 22.

Interactive mechanism for nucleophilic substitution on pyridines



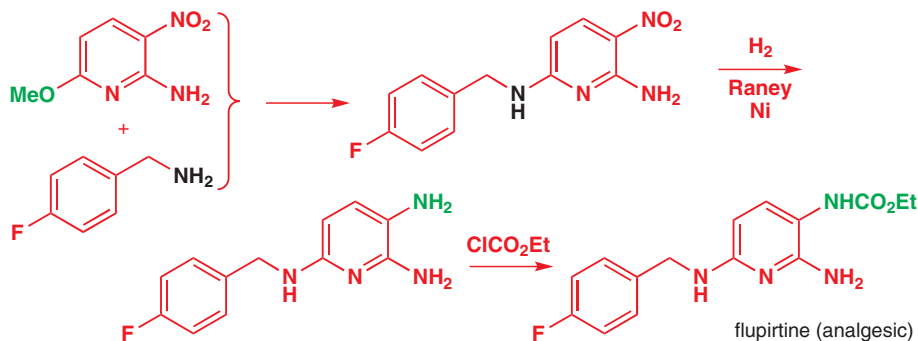
The intermediate anion is stabilized by electronegative nitrogen and by delocalization round the ring. These reactions have some similarity to nucleophilic aromatic substitution (Chapter 22) but are more similar to carbonyl reactions. The intermediate anion is a tetrahedral intermediate that loses the best leaving group to regenerate the stable aromatic system. Nucleophiles such as amines or thiolate anions work well in these reactions.

➡ Note the similarity to nucleophilic substitution on the carbonyl group (Chapter 10).



The leaving group does not have to be as good as chloride in these reactions. Continuing the analogy with carbonyl reactions, 2- and 4-chloropyridines are rather like acid chlorides but we need only use less reactive pyridyl ethers, which react like esters, to make amides. Substitution of a 2-methoxypyridine allows the synthesis of flupirtine.

➡ You will see more of this synthesis later in the chapter.



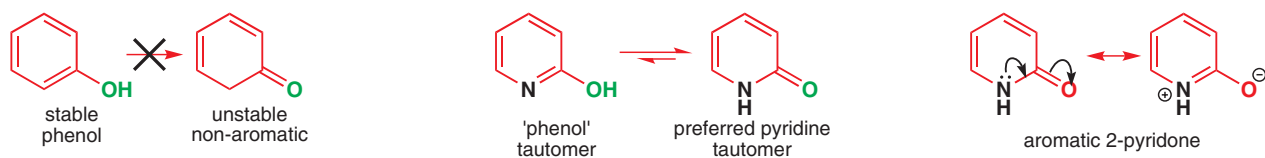
The first step is a nucleophilic aromatic substitution. In the second step the nitro group is reduced to an amino group without any effect on the pyridine ring—another piece of evidence for its aromaticity. Finally, the one amino group whose lone pair is not delocalized onto the pyridine N is acylated in the presence of two others.

Pyridones are good substrates for nucleophilic substitution

The starting materials for these nucleophilic substitutions (2- and 4-chloro- or methoxypyridines) are themselves made by nucleophilic substitution on *pyridones*. If you were asked to propose how 2-methoxypyridine might be made, you would probably suggest, by analogy with the corresponding benzene compound, alkylation of a phenol. Let us look at this in detail.

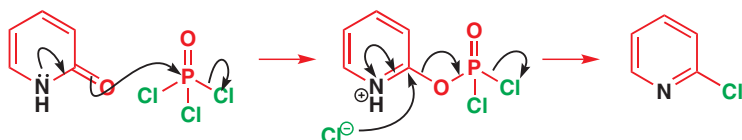


The starting material for this reaction is a 2-hydroxypyridine that can tautomerize to an amide-like structure known as a pyridone by the shift of the acidic proton from oxygen to nitrogen. In the phenol series there is no doubt about which structure will be stable as the ketone is not aromatic; for the pyridine both structures are aromatic.

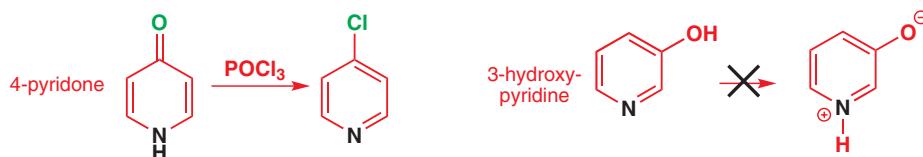


In fact, 2-hydroxypyridine prefers to exist as the 'amide' because that has the advantage of a strong C=O bond and is still aromatic. There are two electrons in each of the C=C double bonds and two also in the lone pair of electrons on the trigonal nitrogen atom of the amide. Delocalization of the lone pair in typical amide style makes the point clearer.

Pyridones are easy to prepare (see Chapter 30) and can be alkylated on oxygen as predicted by their structure. A more important reaction is the direct conversion to chloropyridines with POCl₃. The reaction starts by attack of the oxygen atom at phosphorus to create a leaving group, followed by aromatic nucleophilic substitution. The overall effect is very similar to acyl chloride formation from a carboxylic acid (Chapter 10).



The same reaction occurs with 4-pyridone, which is also delocalized in the same way and exists in the 'amide' form, but not with 3-hydroxypyridine, which exists in the 'phenol' form. Its only tautomer is a zwitterion but the pyridine nitrogen is too weak to remove a proton from the hydroxyl group.

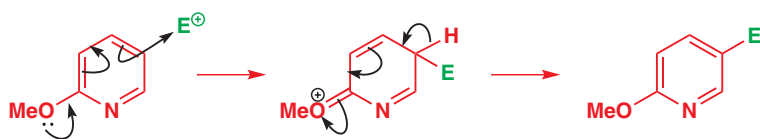


● Pyridines undergo nucleophilic substitution

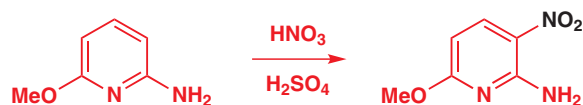
Pyridines can undergo *electrophilic* substitution only if they are activated by electron-donating substituents (see next section) but they readily undergo *nucleophilic* substitution without any activation other than the ring nitrogen atom.

Activated pyridines will do electrophilic aromatic substitution

Useful electrophilic substitutions occur only on pyridines having electron-donating substituents such as NH₂ or OMe. These activate benzene rings too (Chapter 21) but here their help is vital. They supply a non-bonding pair of electrons that raises the energy of the HOMO and carries out the reaction. Simple amino- or methoxypyridines react reasonably well *ortho* and *para* to the activating group. These reactions happen in spite of the molecule being a pyridine, not because of it.



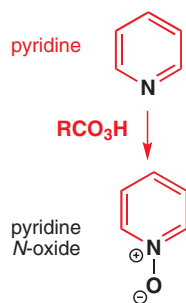
A practical example occurs in the manufacture of the analgesic flupirtine where a doubly activated pyridine having both MeO and NH₂ groups is nitrated just as if it were a benzene ring. The nitro group goes in *ortho* to the amino group and *para* to the methoxy group. The activation is evidently enough to compensate for the molecule being almost entirely protonated under the conditions of the reaction.



Interactive tautomerism between 2-hydroxypyridine and pyridone

➡ This is the starting material for the flupirtine synthesis on p. 728.

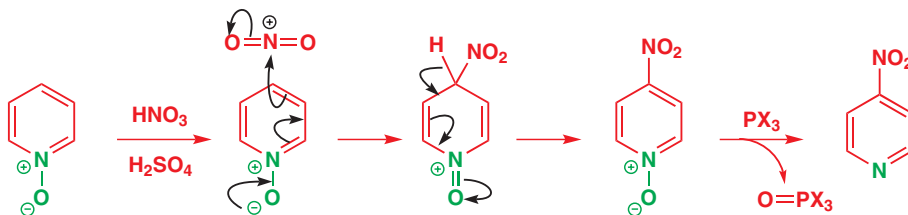
Pyridine *N*-oxides are reactive towards both electrophilic and nucleophilic substitution



Interactive structure of pyridine *N*-oxide

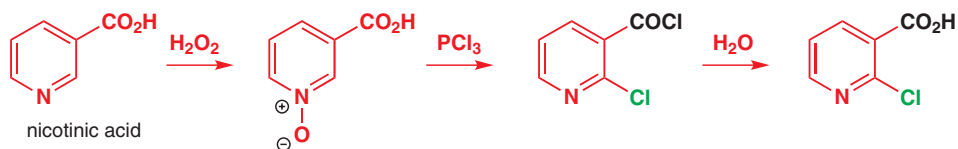
This is all very well if the molecule has such activating groups, but supposing it doesn't? How are we to nitrate pyridine itself? The answer involves an ingenious trick. We need to activate the ring with an electron-rich substituent that can later be removed and we also need to stop the nitrogen atom reacting with the electrophile. All of this can be done with a single atom!

Because the nitrogen atom is nucleophilic, pyridine can be oxidized to pyridine *N*-oxide with reagents such as *m*-CPBA or just H_2O_2 in acetic acid. These *N*-oxides are stable dipolar species with the electrons on oxygen delocalized round the pyridine ring, raising the HOMO of the molecule. Reaction with electrophiles occurs at the 2- (*ortho*) and 4- (*para*) positions, chiefly at the 4-position to keep away from positively charged nitrogen.

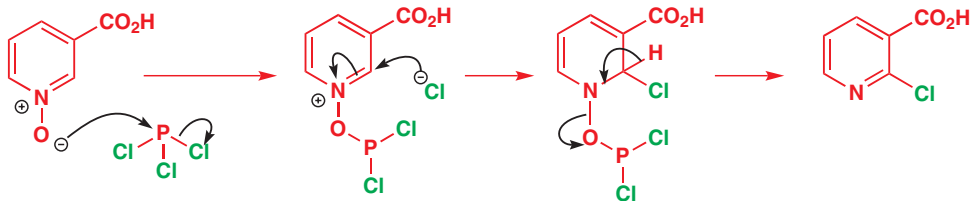


Now the oxide must be removed and this is best done with trivalent phosphorus compounds such as $(\text{MeO})_3\text{P}$ or PCl_3 . The phosphorus atom detaches the oxygen atom in a single step to form the very stable $\text{P}=\text{O}$ double bond. In this reaction the phosphorus atom is acting as both a nucleophile and an electrophile, but mainly as an electrophile since PCl_3 is more reactive here than $(\text{MeO})_3\text{P}$.

The same activation that allowed simple electrophilic substitution—oxidation to the *N*-oxide—can also allow a useful *nucleophilic* substitution. The positive nitrogen atom encourages nucleophilic attack and the oxygen atom can be turned into a leaving group with PCl_3 . Our example is nicotinic acid, whose biological importance we will discuss in Chapter 42.

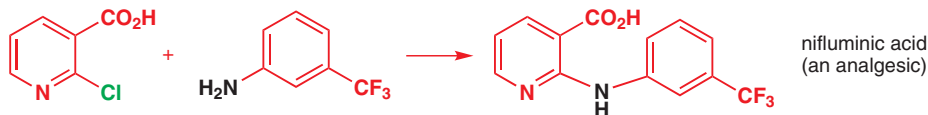


The *N*-oxide reacts with PCl_3 through oxygen and the chloride ion released in this reaction adds to the most electrophilic position between the two electron-withdrawing groups. Now a simple elimination restores aromaticity and gives a product looking as though it results from chlorination rather than nucleophilic attack.



Interactive mechanism for nucleophilic substitution on pyridine *N*-oxide

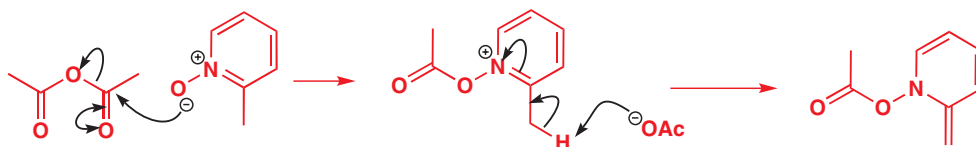
The reagent PCl_3 also converts the carboxylic acid to the acyl chloride, which is hydrolysed back again in the last step. This is a useful sequence because the chlorine atom has been introduced into the 2-position, from which it may in turn be displaced by, for example, amines.



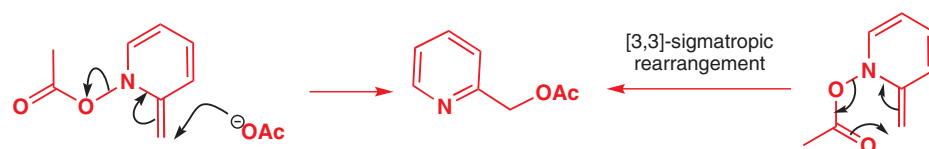
● Pyridine *N*-oxides

Pyridine *N*-oxides are useful for both electrophilic and nucleophilic substitutions on the same carbon atoms (2-, 4-, and 6-) in the ring.

Nucleophilic addition at an even more distant site is possible on reaction with acid anhydrides if there is an alkyl group in the 2-position. Acylation occurs on oxygen as in the last reaction but then a proton is lost from the side chain to give an uncharged intermediate.



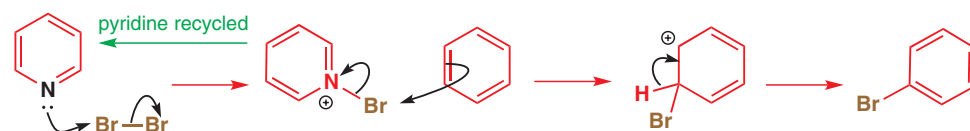
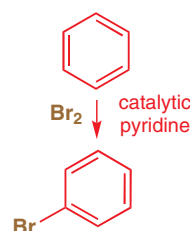
This compound rearranges with migration of the acetate group to the side chain and the restoration of aromaticity. This may be an ionic reaction or a type of rearrangement that you will learn to call a [3,3]-sigmatropic rearrangement (Chapter 35).



Pyridine as a catalyst and reagent

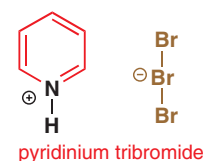
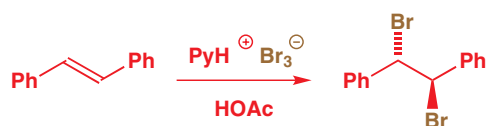
Since pyridine is abundant and cheap and has an extremely rich chemistry, it is not surprising that it has many applications. One of the simplest ways to brominate benzenes is not to bother with the Lewis acid catalysts recommended in Chapter 21 but just to add liquid bromine to the aromatic compound in the presence of a small amount of pyridine. Only about one mole per cent is needed and even then the reaction has to be cooled to stop it getting out of hand.

As we have seen, pyridine attacks electrophiles through its nitrogen atom. This produces the reactive species, the *N*-bromo-pyridinium ion, which is attacked by the benzene. Pyridine is a better nucleophile than benzene and a better leaving group than bromide. This is another example of **nucleophilic catalysis**.



➡ Nucleophilic catalysis is discussed on p. 200.

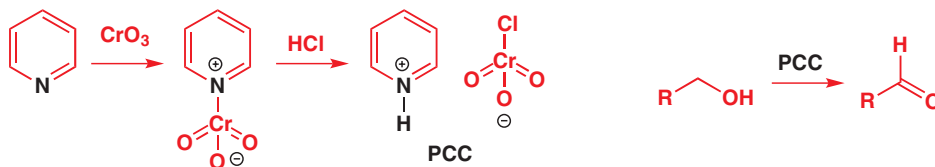
Another way to use pyridine in brominations is to make a stable crystalline compound to replace the dangerous liquid bromine. This compound, known by names such as pyridinium tribromide, is simply a salt of pyridine with the anion Br_3^- . It can be used to brominate reactive compounds such as alkenes (Chapter 19).



Both of these methods depend on the lack of reactivity of pyridine's π system towards electrophiles such as bromine. Notice that, in the first case, both benzene and pyridine are present together. The pyridine attacks bromine only through nitrogen (and reversibly at that) and never through carbon.

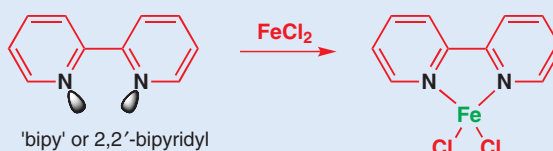
Oxidation of alcohols is normally carried out with Cr(VI) reagents (Chapter 23) but these, like the Jones' reagent ($\text{Na}_2\text{Cr}_2\text{O}_7$ in sulfuric acid), are usually acidic. Some pyridine complexes

of Cr(VI) compounds solve this problem by having the pyridinium ion (pK_a 5) as the only acid. The two most famous are PDC (pyridinium dichromate) and PCC (pyridinium chlorochromate). Pyridine forms a complex with CrO_3 but this is liable to burst into flames. Treatment with HCl gives PCC, which is much less dangerous. PCC is particularly useful in the oxidation of primary alcohols to aldehydes as over-oxidation is avoided in the only slightly acidic conditions (Chapter 23).

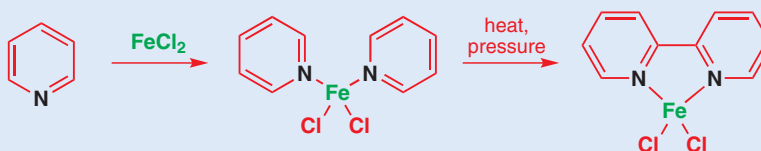


Bipyridyl (bipy)

The ability of pyridine to form metal complexes is greatly enhanced in a dimer—the famous ligand ‘bipy’ or 2,2′-bipyridyl. It is bidentate and because of its ‘bite’ it is a good ligand for many transition metals, with a partiality for Fe(II).

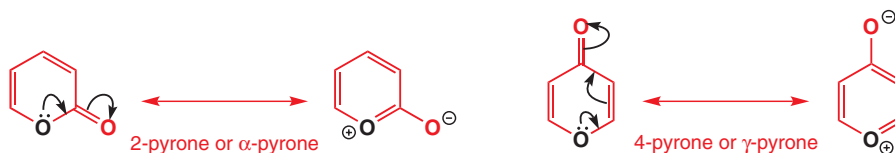


It looks like a rather difficult job to persuade two pyridine rings to join together in this way to form bipy. It is indeed very difficult unless you make things easier by using a reagent that favours the product. And what better than Fe(II) to do the job? Bipy is manufactured by treating pyridine with $\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$ at high temperatures and high pressures. Only a small proportion of the pyridine is converted to the Fe(II) complex of bipy (about 5%) but the remaining pyridine goes back in the next reaction. This is probably a radical process (Chapter 37) within the coordination sphere of Fe(II).

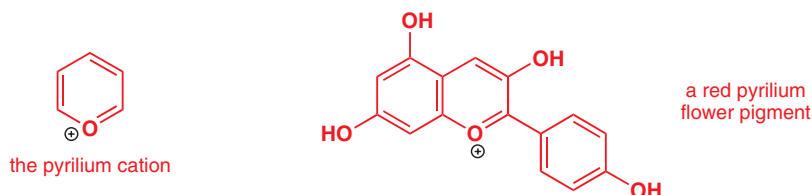


Six-membered aromatic heterocycles can have oxygen in the ring

Although pyridine is overwhelmingly the most important of the six-membered aromatic heterocycles, there are oxygen heterocycles, pyrones, that resemble the pyridones. The pyrones are aromatic, although α -pyrone is rather unstable.

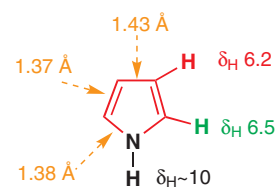


The pyrylium salts are stable aromatic cations and are responsible as metal complexes for some flower colours. Heterocycles with six-membered rings based on other elements (for example, P) do exist but they are outside the scope of this book.



Five-membered aromatic heterocycles are good at electrophilic substitution

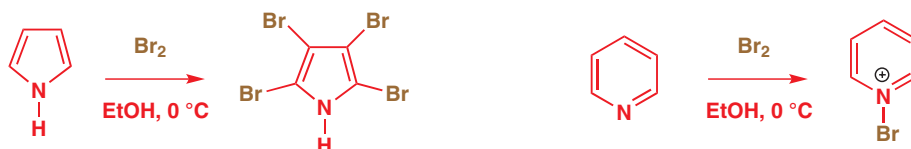
Just about everything is the other way round with pyrrole. Electrophilic substitution is much easier than it is with benzene—almost too easy in fact—while nucleophilic substitution is more difficult. Pyrrole is not a base nor can it be converted to an *N*-oxide. We need to find out why this is. The big difference is that the nitrogen lone pair is delocalized round the ring. The NMR spectrum suggests that all the positions in the ring are about equally electron-rich with chemical shifts about 1 ppm smaller than those of benzene. The ring is flat and the bond lengths are very similar, although the bond opposite the nitrogen atom is a bit longer than the others.



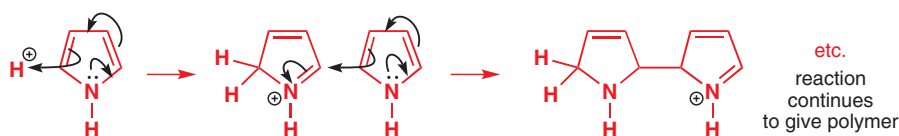
Interactive structure of pyrrole

The delocalization of the lone pair can be drawn equally well to any ring atom because of the five-membered ring and we shall soon see the consequences of this. All the delocalization pushes electrons from the nitrogen atom into the ring and we expect the ring to be electron-rich at the expense of the nitrogen atom. The HOMO should go up in energy and the ring become more nucleophilic.

An obvious consequence of this delocalization is the decreased basicity of the nitrogen atom and the increased acidity of the NH group. In fact, the pK_a of pyrrole acting as a base is about -4 , and protonation occurs at carbon below pH -4 . By contrast, the NH proton (pK_a 16.5) can be removed by much weaker bases than those that can remove protons on normal secondary amines. The nucleophilic nature of the ring means that pyrrole is attacked readily by electrophiles. Reaction with bromine requires no Lewis acid and leads to substitution (confirming the aromaticity of pyrrole) at all four free positions. Contrast pyridine's reactivity with bromine (p. 731): it reacts just once, at nitrogen.



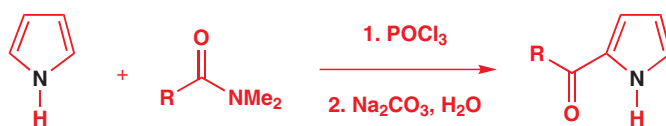
This is a fine reaction in its way, but we don't usually want four bromine atoms in a molecule so one problem with pyrrole is to control the reaction to give only monosubstitution. Another problem is that strong acids cannot be used. Although protonation does not occur at nitrogen, it does occur at carbon and the protonated pyrrole then adds another molecule like this.



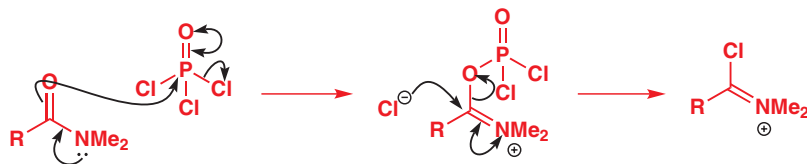
● Pyrrole polymerizes!

Strong acids, those such as H_2SO_4 with a pK_a of less than -4 , cannot be used without polymerization of pyrrole.

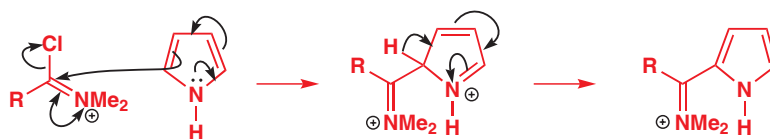
Some reactions can be controlled to give good yields of monosubstituted products. One is the Vilsmeier reaction, in which a combination of an *N,N*-dimethylamide and $POCl_3$ is used to make a carbon electrophile in the absence of strong acid or Lewis acid. It is a substitute for the Friedel–Crafts acylation, and works with aromatic compounds at the more reactive end of the scale (where pyrrole is).



In the first step, the amide reacts with POCl_3 , which makes off with the amide oxygen atom and replaces it with chlorine. This process would be very unfavourable but for the formation of the strong $\text{P}=\text{O}$ bond, and is the direct analogy of the chloropyridine-forming reaction you have just seen.



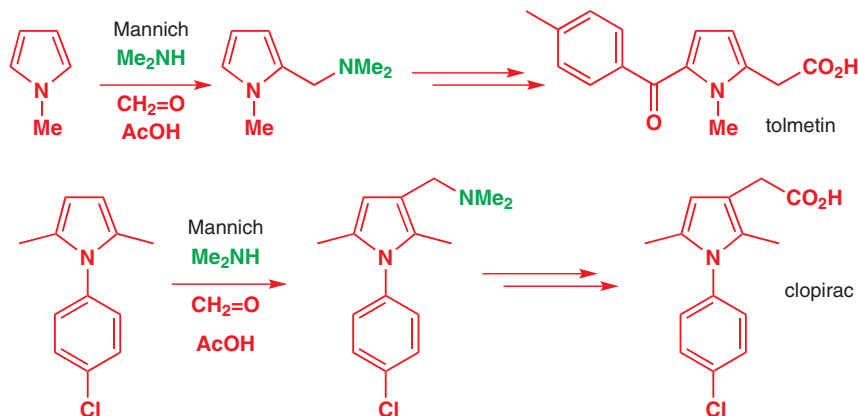
The product from this first step is an iminium cation that reacts with pyrrole to give a more stable iminium salt. The extra stability comes from the conjugation between the pyrrole nitrogen and the iminium group. The work-up with aqueous Na_2CO_3 hydrolyses the imine salt and removes any acid formed. This method is particularly useful because it works well with Me_2NCHO (DMF) to add a formyl (CHO) group. This is difficult to do with a conventional Friedel–Crafts reaction.



Interactive mechanism for Vilsmeier reaction of pyrrole

➡ Remind yourself of the Mannich reaction on p. 621 of Chapter 26.

You may have noticed that the reaction occurred only at the 2-position on pyrrole. Although all positions react with reagents like bromine, most reagents go for the 2- (or 5-) position and attack the 3- (or 4-) position only if the 2- and 5-positions are blocked. A good example is the Mannich reaction. In these two examples *N*-methylpyrrole reacts cleanly at the 2-position while the other pyrrole with both 2- and 5-positions blocked by methyl groups reacts cleanly at the 3-position. These reactions are used in the manufacture of the non-steroidal anti-inflammatory compounds tolmetin and clopirac.

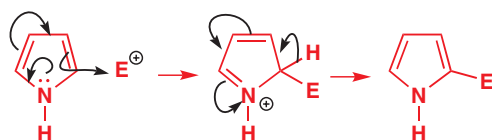


Interactive mechanism for the Mannich reaction on pyrrole

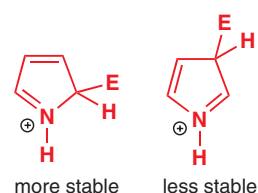
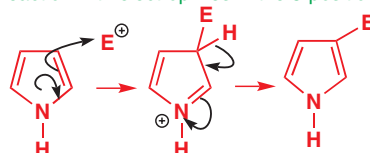
Now we need an explanation. The mechanisms for both 2- and 3-substitutions look good and we will draw both, using a generalized E^+ as the electrophile. Both mechanisms can occur very readily. Reaction in the 2-position is somewhat better than in the 3-position but the difference is small. Substitution is favoured at *all* positions. Calculations show that the HOMO of pyrrole does indeed have a larger coefficient in the 2-position, and one way to explain this result is to look at the structure of the intermediates. The intermediate from attack at the

2-position has a linear conjugated system. In both intermediates the two double bonds are, of course, conjugated with each other, but only in the first intermediate are both double bonds conjugated with N^+ . The second intermediate is 'cross-conjugated', while the first has a more stable linear conjugated system.

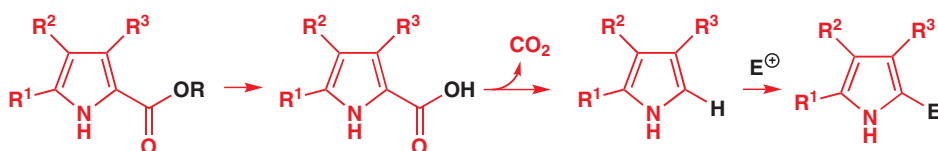
reaction with electrophiles in the 2-position



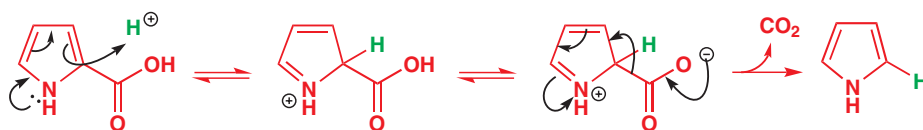
reaction with electrophiles in the 3-position



Since electrophilic substitution on pyrroles occurs so easily, it can be useful to block substitution with a removable substituent. This is usually done with an ester group. Hydrolysis of the ester (this is particularly easy with *t*-butyl esters—see Chapter 23) releases the carboxylic acid, which decarboxylates on heating. There is no doubt that the final electrophilic substitution must occur at C2.



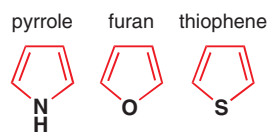
The decarboxylation is a general reaction of pyrroles: it's a kind of reverse Friedel–Crafts reaction in which the electrophile is a proton (provided by the carboxylic acid itself) and the leaving group is carbon dioxide. The protonation may occur anywhere but it leads to reaction only if it occurs where there is a CO_2H group.



Furan and thiophene are oxygen and sulfur analogues of pyrrole

The other simple five-membered heterocycles are furan, with an oxygen atom instead of nitrogen, and thiophene, with a sulfur atom. They also undergo electrophilic aromatic substitution very readily, although not so readily as pyrrole. Nitrogen is the most powerful electron donor of the three, oxygen the next, and sulfur the least. Thiophene is very similar to benzene in reactivity.

Thiophene is the least reactive of the three because the p orbital of the lone pair of electrons on sulfur that conjugates with the ring is a 3p orbital rather than the 2p orbital of N or O, so overlap with the 2p orbitals on carbon is less good. Both furan and thiophene undergo more or less normal Friedel–Crafts reactions, although the less reactive anhydrides (here acetic anhydride, Ac_2O) are used instead of acid chlorides, and weaker Lewis acids than $AlCl_3$ are preferred.



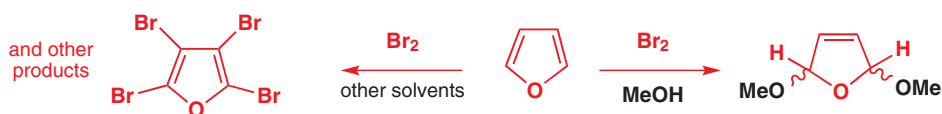
Notice that the regioselectivity is the same as it was with pyrrole—the 2-position is more reactive than the 3-position in both cases. The product ketones are less reactive towards electrophiles than the starting heterocycles and deactivated furans can even be nitrated

with the reagents used for benzene derivatives. Notice that reaction has occurred at the 5-position in spite of the presence of the ketone. The preference for 2- and 5-substitution is quite marked.



Electrophilic addition may be preferred to substitution with furan

So far, thiophenes and furans look much the same as pyrrole but there are other reactions in which they behave quite differently and we shall now concentrate on those. Furan is less aromatic than pyrrole, and if there is the prospect of forming stable bonds such as C–O single bonds by addition, this may be preferred to substitution. A famous example is the reaction of furan with bromine in methanol. In non-hydroxylic solvents, polybromination occurs as expected, but in MeOH no bromine is added at all!



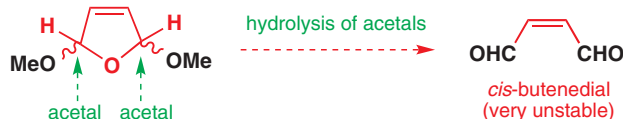
Bromination must start in the usual way, but a molecule of methanol captures the first formed cation in a 1,4-addition to furan.



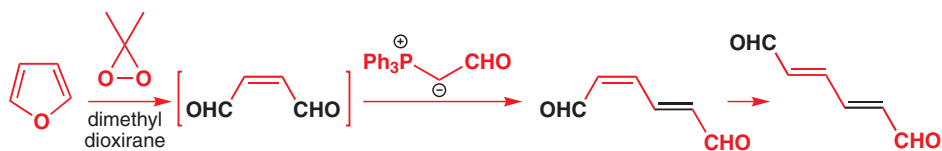
The bromine atom that was originally added is now pushed out by the furan oxygen atom to make a relatively stable conjugated oxonium ion, which adds a second molecule of methanol.



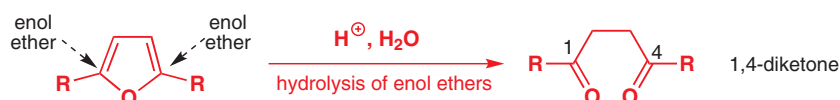
This product conceals an interesting molecule. At each side of the ring we have an acetal, and if we were to hydrolyse the acetals, we would have 'maleic dialdehyde' (*cis*-butenedial)—a molecule that is too unstable to be isolated. The furan derivative may be used in its place.



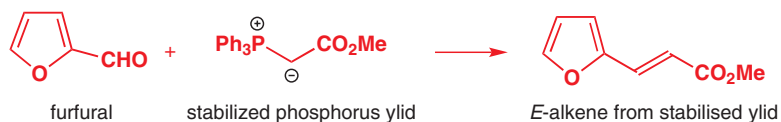
The same 1,4-dialdehyde can be made by oxidizing furan with the mild oxidizing agent dimethyldioxirane, which you met on p. 432. In this sequence, it is trapped in a Wittig reaction to give an *E,Z*-diene, which is easily isomerized to *E,E*.



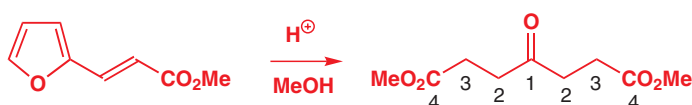
We can extend this idea of furan being the origin of 1,4-dicarbonyl compounds if we consider that furan is, in fact, an enol ether on both sides of the ring. If these enol ethers were hydrolysed we would get a 1,4-diketone.



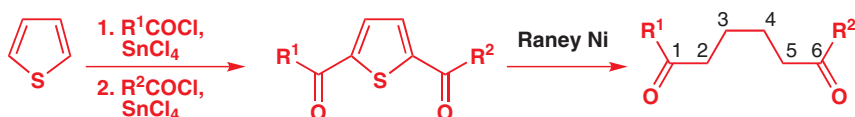
This time the arrow is solid, not dotted, because this reaction really happens. You will discover in the next chapter that furans can also be made from 1,4-diketones so this whole process is reversible. The example we are choosing has other features worth noting. The cheapest starting material containing a furan is furan-2-aldehyde or 'furfural', a by-product of breakfast cereal manufacture. Here it reacts in a typical Wittig process with a stabilized ylid.



Now comes the interesting step: treatment of this furan with acidic methanol gives a white crystalline compound having two 1,4-dicarbonyl relationships. You might like to try and draw a mechanism for this reaction.



The thiophene ring can also be opened up, but in a very different way. Reductive removal of the sulfur atom with Raney nickel reduces not only the C–S bonds but also the double bonds in the ring and the four carbons in the ring form a saturated alkyl chain. If the reduction follows two Friedel–Crafts reactions on thiophene the product is a 1,6-diketone instead of the 1,4-diketones from furan. Thiophene is well behaved in Friedel–Crafts acylations, and reaction occurs at the 2- and 5-positions unless these are blocked.

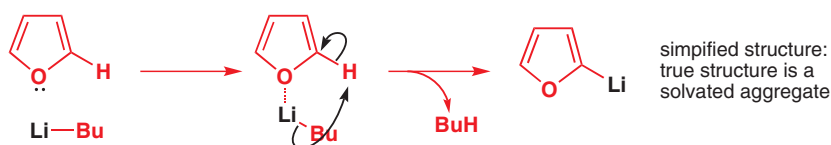


Lithiation of thiophenes and furans

A reaction that furans and thiophenes do particularly well and that fits well with these last two reactions is metallation, particularly lithiation, of a C–H group next to the heteroatom. Metallation of benzene rings (Chapter 24) is carried out by lithium–halogen (Br or I) exchange—a method that works well for heterocycles too as we will see later with pyridine—or by directed (*ortho*) lithiation of a C–H group next to an activating group such as OMe. With thiophene and furan, the heteroatom in the ring provides the necessary activation.



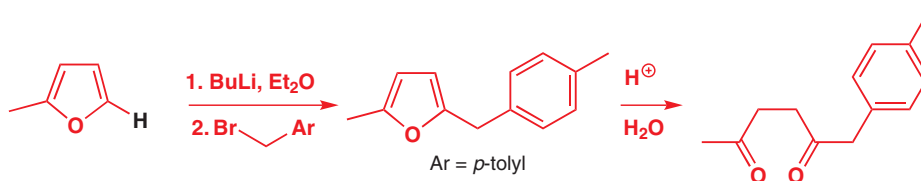
Activation is by coordination of O or S to Li followed by proton removal by the butyl group—the by-product is gaseous butane. These lithium compounds have a carbon–lithium σ bond and are soluble in organic solvents. We shall represent them very simply, but in fact they are typically dimers or more complex aggregates, with the coordination sphere of Li completed by THF molecules.



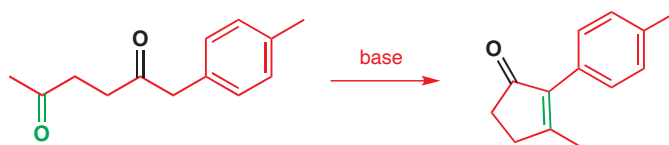
➡ We explained some of the challenges in making 1,4-difunctionalized compounds in Chapter 28.

➡ Raney nickel was introduced in Chapter 23, p. 537.

These lithium compounds are very reactive and will combine with most electrophiles—in this example the organolithium is alkylated by a benzylic halide. Treatment with aqueous acid gives the 1,4-diketone by hydrolysis of the two enol ethers.



Treatment of this diketone with *anhydrous* acid would cause recyclization to the same furan (see Chapter 30) but it can alternatively be cyclized in base by an intramolecular aldol reaction (Chapter 26) to give a cyclopentenone.

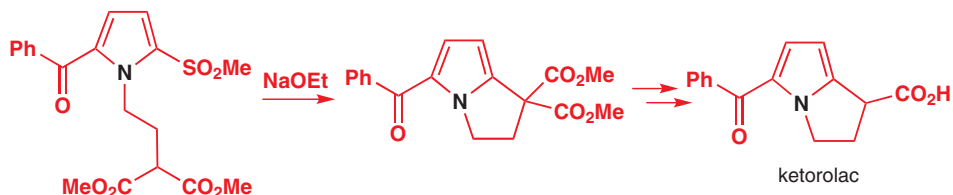


This completes our exploration of chemistry special to thiophene and furan, and we now return to all three heterocycles (pyrrole in particular) and look at *nucleophilic* substitution.

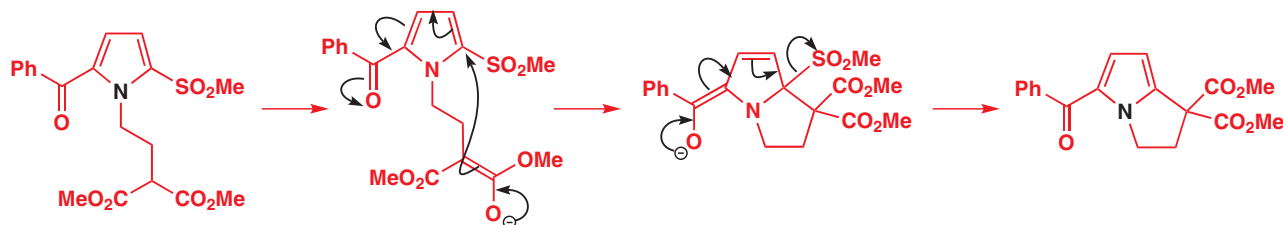
More reactions of five-membered heterocycles

Nucleophilic substitution requires an activating group

Nucleophilic substitution is a relatively rare reaction with pyrrole, thiophene, or furan and requires an activating group such as nitro, carbonyl, or sulfonyl, just as it does with benzene (Chapter 22). This intramolecular example is used to make the painkiller ketorolac.



The nucleophile is a stable enolate and the leaving group is a sulfonate anion. An intermediate must be formed in which the negative charge is delocalized onto the carbonyl group on the ring, just as you saw in the benzene ring examples in Chapter 22. Attack occurs at the 2-position because the leaving group is there and because the negative charge can be delocalized onto the ketone from that position.

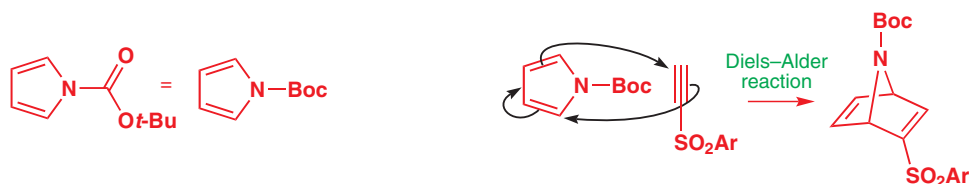


Five-membered heterocycles act as dienes in Diels–Alder reactions

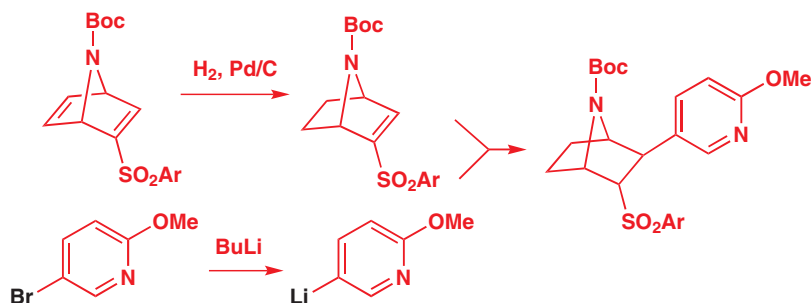
All of the reactions of pyrrole, furan, and thiophene we have discussed so far have been variations on reactions of benzene. But heterocycles also do reactions totally unlike those of benzene and we are now going to explore two of them.

The first is a reaction you will meet in detail in Chapter 34. It is known as the Diels–Alder reaction, and although it has a number of subtleties we will not discuss here, it has a simple cyclic mechanism in which six electrons (three curly arrows) move around to form a new six-membered ring.

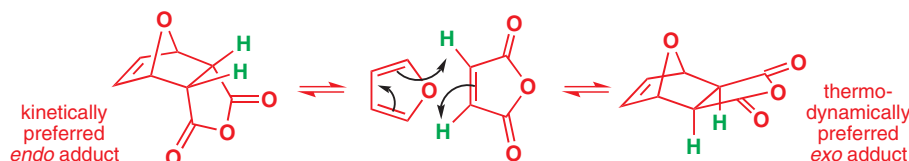
Here is an example with the Boc derivative of pyrrole. The electron-deficient Boc group makes pyrrole less nucleophilic and promotes the Diels–Alder reaction with an alkynyl sulfone. Benzene, and even many other heterocycles, will not do this sort of reaction.



The product is a useful intermediate in the synthesis of the analgesic epibatidine. Selective reduction of the non-conjugated double bond is followed by addition of a pyridine nucleophile (a lithium derivative can be prepared from a bromopyridine) to the vinyl sulfone.



Furan is particularly good at Diels–Alder reactions but it gives the thermodynamic product, the *exo* adduct, because with this aromatic diene the reaction is reversible.



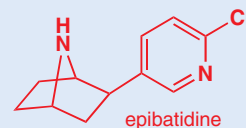
Aromaticity prevents thiophene taking part in Diels–Alder reactions, but oxidation to the sulfone destroys the aromaticity because both lone pairs become involved in bonds to oxygen. The sulfone is unstable and reacts with itself but will also do Diels–Alder reactions. With an alkyne, loss of SO₂ gives a substituted benzene derivative.



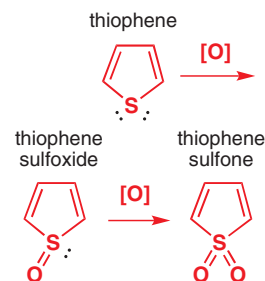
Similar reactions occur with α -pyrones. These are also rather unstable and barely aromatic and they react with alkynes by Diels–Alder reactions followed by reverse Diels–Alder reactions to give benzene derivatives with the loss of CO₂.

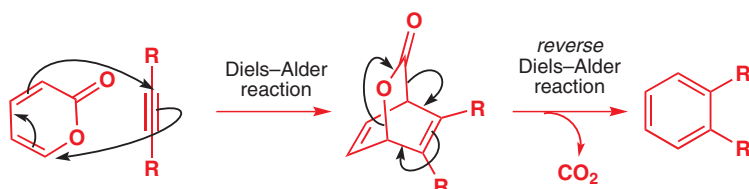
➡ The Boc protecting group is discussed in Chapter 23, p. 558.

Epibatidine was discovered in the skin of Ecuadoran frogs in 1992. It is an exceptionally powerful analgesic and works by a different mechanism from that of morphine so there is hope that it will not be addictive. The compound can now be synthesized so there is no need to kill the frogs to get it—indeed, they are a protected species.

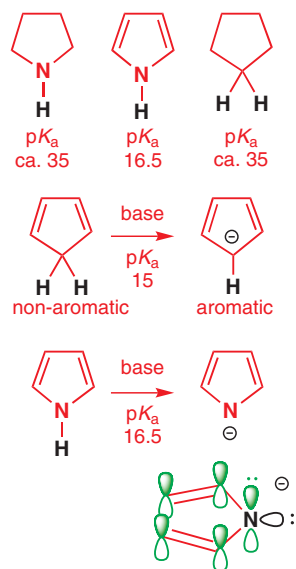


➡ *Endo* and *exo* Diels–Alder adducts are explained in Chapter 34.



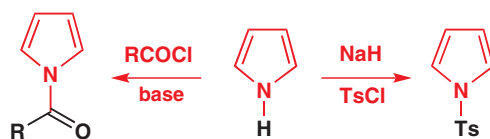


Nitrogen anions can be easily made from pyrrole



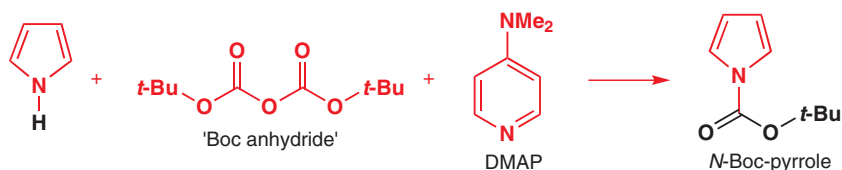
Pyrrole is much more acidic than comparable saturated amines. The pK_a of pyrrolidine is about 35, but pyrrole has a pK_a of 16.5, making it some 10^{23} times more acidic! Pyrrole is about as acidic as a typical alcohol so bases stronger than alkoxides will convert it to its anion. We should not be too surprised at this as the corresponding hydrocarbon, cyclopentadiene, is also extremely acidic, with a pK_a of 15. The reason is that the anions are aromatic with six delocalized π electrons. The effect is much greater for cyclopentadiene because the hydrocarbon is not aromatic and much less for pyrrole because it is already aromatic and has less to gain.

In all of the reactions of pyrrole that we have so far seen, new groups have added to the carbon atoms of the ring. The anion of pyrrole is useful because it reacts at nitrogen. The nitrogen atom has two lone pairs of electrons in the anion: one is delocalized around the ring but the other is localized in an sp^2 orbital on nitrogen. This high-energy pair is the new HOMO and this is where the molecule reacts. *N*-acylated derivatives in general can be made in this way. A commonly used base is sodium hydride (NaH) but weaker bases produce enough anion for reaction to occur.



● Anions of pyrroles react with electrophiles at the *nitrogen* atom.

This is how the *N*-Boc pyrrole was made for use in the synthesis of epibatidine on p. 739. The base used was the pyridine derivative DMAP, which you met earlier in the chapter (p. 726). Its conjugate acid has a pK_a of 9.7 and so produces small, equilibrating amounts of the anion as well as acting as a nucleophilic catalyst. 'Boc anhydride' is used as the acylating agent.



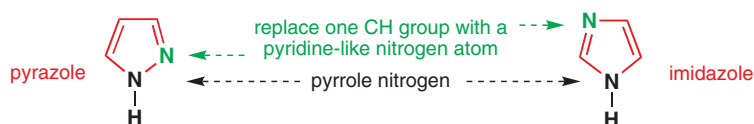
Anion formation is important in the next main section of this chapter, which is about what happens when we insert more nitrogen atoms into the pyrrole ring.

Five-membered rings with two or more nitrogen atoms

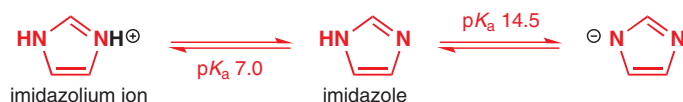
Imidazole

At the beginning of this chapter we imagined adding more nitrogen atoms to the pyrrole ring and noticed then that there were two compounds with two nitrogen atoms: pyrazole and imidazole.

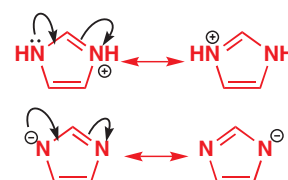
➡ DMAP's pK_a of 9.7 is between those of pyridine (5.5) and tertiary alkyl amines (ca. 10) but is much closer to the latter.



Only one nitrogen atom in a five-membered ring can contribute two electrons to the aromatic sextet. The other replaces a CH group, has no hydrogen, and is like the nitrogen atom in pyridine. The black nitrogens are the pyrrole-like nitrogens; the green ones are pyridine-like. The lone pairs on the black nitrogens are delocalized round the ring; those on the green nitrogens are localized in sp^2 orbitals on nitrogen. We can expect these compounds to have properties intermediate between those of pyrrole and pyridine. Imidazole is a stronger base than either pyrrole or pyridine—the imidazolium ion has a pK_a of almost exactly 7, meaning that it is 50% protonated in neutral water. Imidazole is also more acidic than pyrrole, with a pK_a of 14.5.

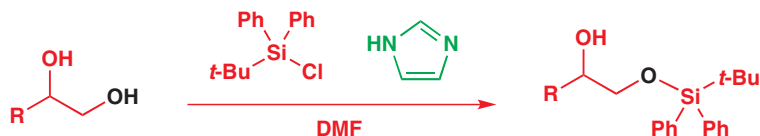


These curious results are a consequence of the 1,3 relationship between the two nitrogen atoms. Both the (protonated) cation and the (deprotonated) anion share the charge equally between the two nitrogen atoms—they are perfectly symmetrical and unusually stable. Another way to look at the basicity of imidazole would be to say that both nitrogen atoms can act at once on the proton being attacked. It has to be the pyridine-like nitrogen that actually captures the proton but the pyrrole nitrogen can help by using its delocalized electrons like this:

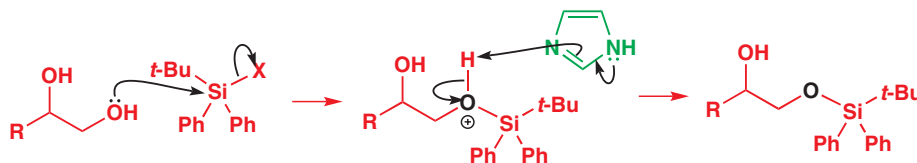


➡ A similar effect accounts for the basicity of DBU, see p. 175.

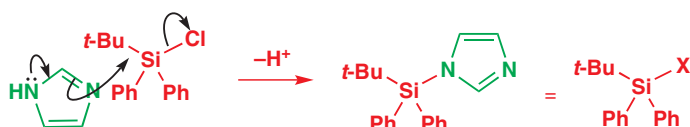
Nature makes use of this property by having imidazole groups attached to proteins in the form of the amino acid histidine and using them as nucleophilic, basic, and acidic catalytic groups in enzyme reactions (this will be discussed in Chapter 42). We use this property in the same way when we add a silyl group to an alcohol. Imidazole is a popular catalyst for these reactions.



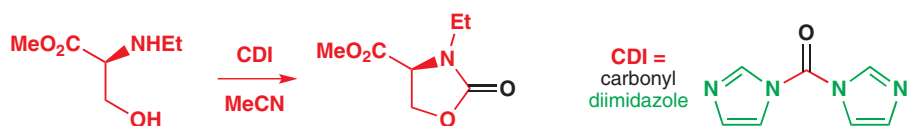
A weakly basic catalyst is needed here because we want to discriminate between the primary and secondary alcohols in the diol. Imidazole is too weak a base to remove protons from an alcohol ($pK_a \sim 16$) but it can remove a proton after the OH group has attacked the silicon atom.



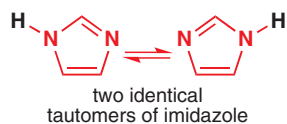
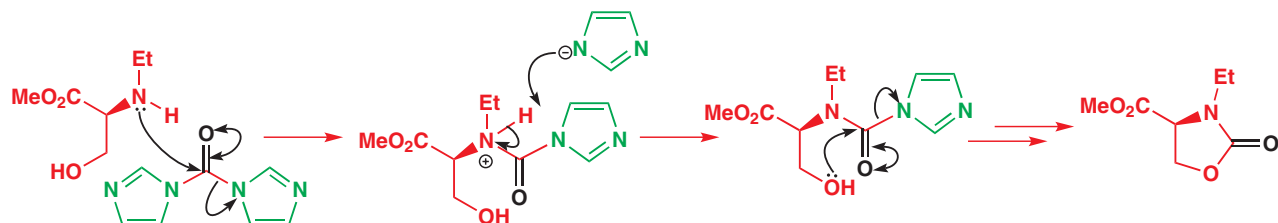
In fact, the imidazole is also a nucleophilic catalyst of this reaction, and the first step is substitution of Cl by imidazole—that is why the leaving group in the last scheme was shown as 'X'. The reaction starts off like this:



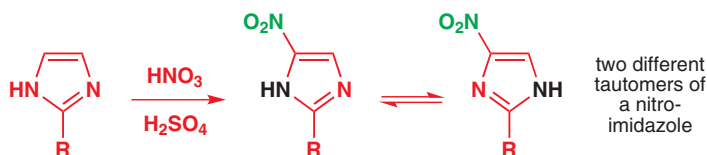
The same idea leads to the use of carbonyl diimidazole (CDI) as a double electrophile when we want to link two nucleophiles together by a carbonyl group. Phosgene (COCl_2) has been used for this but it is appallingly toxic (it was used in the First World War as a poison gas with dreadful effects). CDI is safer and more controlled. In these reactions imidazole acts (twice) as a leaving group.



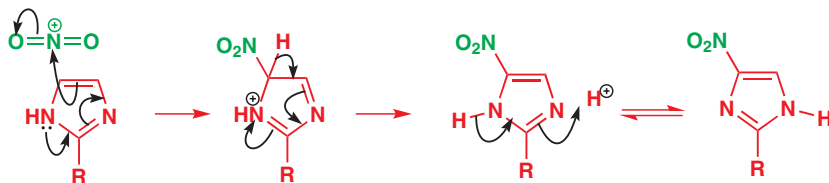
The amino group probably attacks first to displace one imidazole anion, which returns to deprotonate the ammonium salt. The alcohol can then attack intramolecularly, displacing the second imidazole anion, which deprotonates the OH group in its turn. The other product is just two molecules of imidazole.



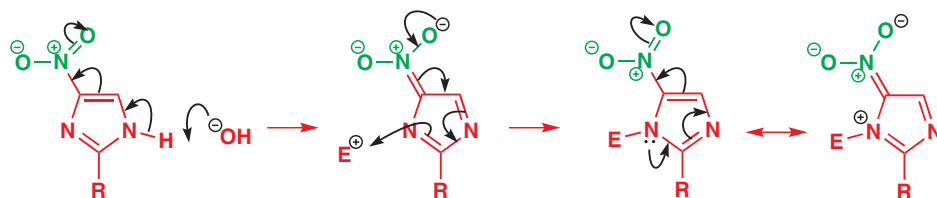
The relationship between the delocalized imidazole anion and imidazole itself is rather like that between an enolate anion and an enol. It will come as no surprise therefore that, like an enol, imidazole tautomerizes rapidly at room temperature in solution. For the parent compound the two tautomers are the same, but with unsymmetrical imidazoles the tautomerism is more interesting. We will explore this question alongside electrophilic aromatic substitution of imidazoles. Imidazoles with a substituent between the two nitrogen atoms (position 2) can be nitrated with the usual reagents and the product consists of a mixture of tautomers.



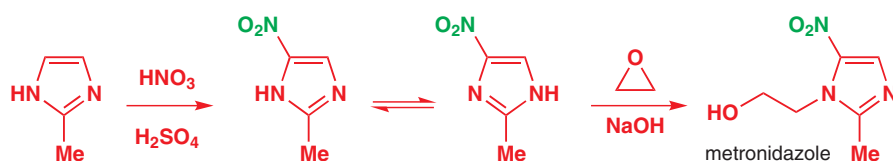
The initial nitration may occur at either of the remaining sites on the ring with the electrons coming from the pyrrole-like nitrogen atom. Tautomerism after nitration gives the mixture. Tautomerism is rapid and the tautomers cannot be separated.



The tautomerism can be stopped by alkylation at one of the nitrogen atoms. If this is done in basic solution, the anion is an intermediate and the alkyl group adds to the nitrogen atom next to the nitro group. Again, it does not matter from which tautomer the anion is derived—there is only one anion delocalized over both nitrogen atoms and the nitro group. One reason for the formation of this isomer is that it has the linear conjugated system between the pyrrole-like nitrogen and the nitro group (see p. 734).

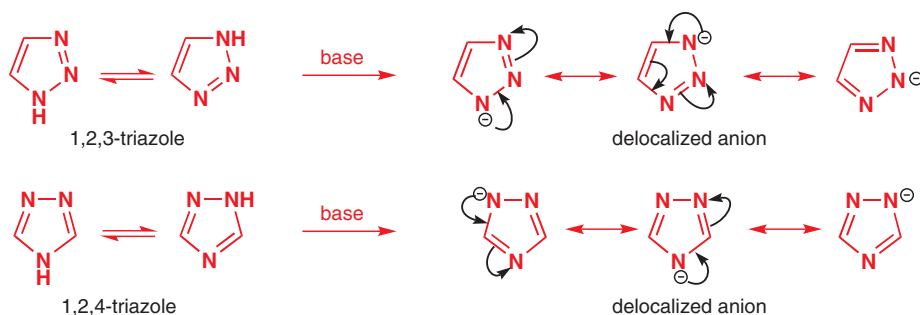


Important medicinal compounds are made in this way. The antiparasitic metronidazole comes from 2-methyl imidazole by nitration and alkylation with an epoxide in base.

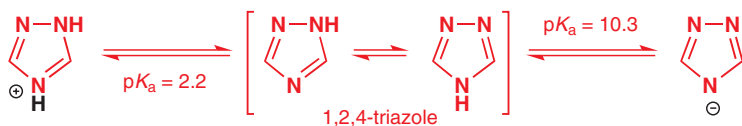


The triazoles

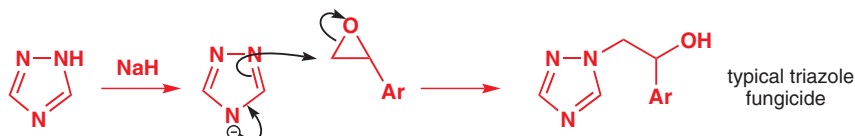
There are two triazoles, and each has one pyrrole-like nitrogen and two pyridine-like nitrogens. Both triazoles have the possibility of tautomerism (in 1,2,3-triazole the tautomers are identical) and both give rise to a single anion.



The 1,2,4-triazole is more important because it is the basis of the best modern agricultural fungicides as well as drugs for fungal diseases in humans. The extra nitrogen atom, inevitably of the pyridine type, makes it more weakly basic than imidazole, but it increases its acidity so that the anion is now easy to make.

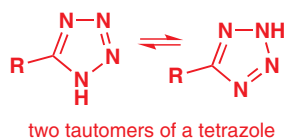
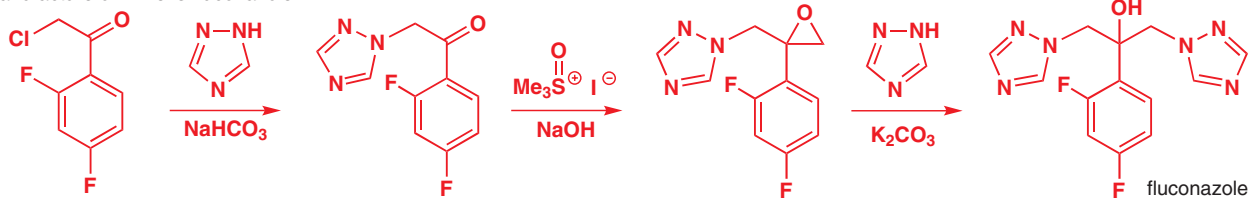


The fungicides are usually made by the addition of the triazole anion to an epoxide or other carbon electrophile. The anion normally reacts at one of the two linked nitrogen atoms (it does not matter which—the product is the same).



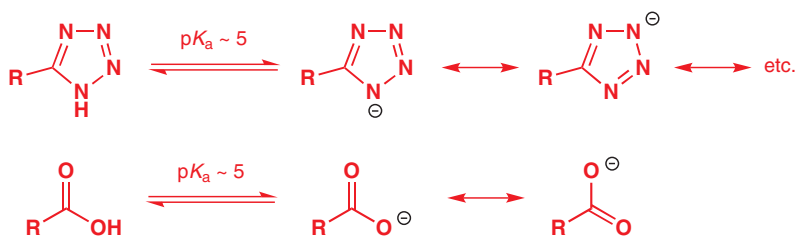
A modern example of an agent used against human fungal infections is Pfizer's fluconazole, which actually contains two triazoles. The first is added as the anion to an α -chloroketone and the second is added to an epoxide made with the sulfur ylid chemistry you met in Chapter 27. Note that weak bases were used to catalyse both of these reactions. Triazole is acidic enough for even NaHCO_3 to produce a small amount of the anion.

manufacture of Pfizer's fluconazole

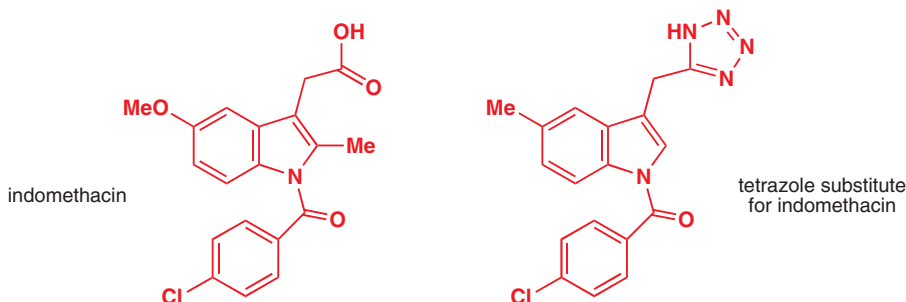


Tetrazole

There is only one isomer of tetrazole or of C-substituted tetrazoles, as there is only one carbon atom in the ring, although there may be several tautomers. The main interest in tetrazoles is that they are rather acidic: the pK_a for the loss of the NH proton to form an anion is about 5, essentially the same as that of a carboxylic acid. The anion is delocalized over all four nitrogen atoms (as well as the one carbon atom), and four nitrogen atoms do the work of two oxygen atoms.

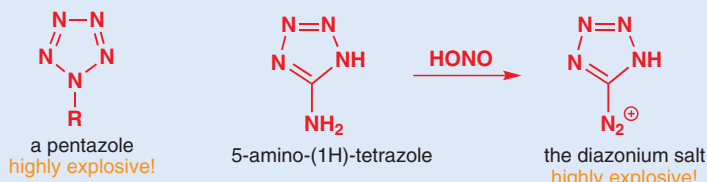


Because tetrazoles have similar acidities to those of carboxylic acids, they have been used in drugs as replacements for the CO_2H unit when the carboxylic acid has unsatisfactory properties for human medicine. A simple example is the anti-arthritis drug indomethacin whose carboxylic acid group may be replaced by a tetrazole with no loss of activity.

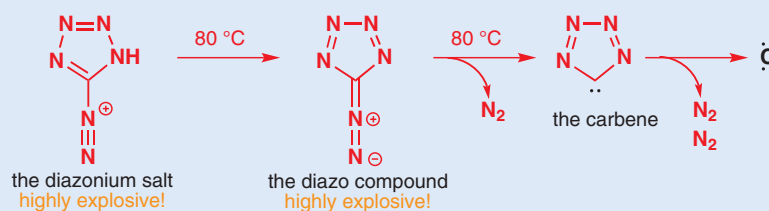


Nitrogen atoms and explosions

Compounds with even two or three nitrogen atoms joined together, such as diazomethane (CH_2N_2) or azides (RN_3), are potentially explosive because they can suddenly give off stable gaseous nitrogen. Compounds with more nitrogen atoms, such as tetrazoles, are likely to be more dangerous and few people have attempted to prepare pentazoles. The limit is reached with diazotetrazole, with the amazing formula CN_6 ! It is made by diazotization of 5-amino-1H-tetrazole, which first gives a diazonium salt.



The diazonium salt is extremely dangerous: 'It should be emphasised that [the diazonium salt] is extremely explosive and should be handled with great care. We recommend that no more than 0.75 mmol be isolated at one time. Ethereal solutions are somewhat more stable but explosions have occurred after standing at $-70\text{ }^{\circ}\text{C}$ for 1 hr.' So much for that, but what about the diazo compound? It is extremely unstable and decomposes to a carbene with loss of one molecule of nitrogen and then loses two more to give...

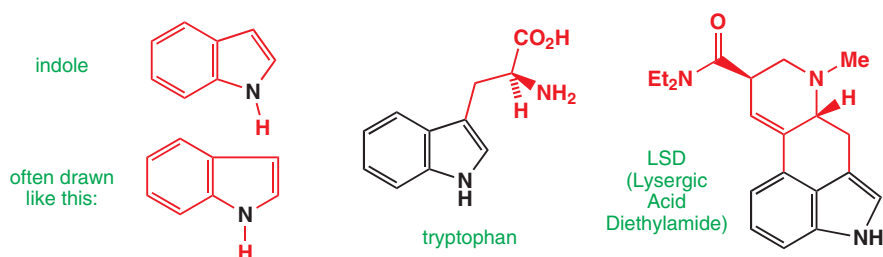


All that is left is a carbon atom and this is one of very few ways to make carbon atoms chemically. The carbon atoms have remarkable reactions and these have been studied briefly, but the hazardous preparation of the starting materials discourages too much research. However, you will see in the next chapter that 1-amino tetrazole is a useful starting material for making an anti-allergic drug.

Benzo-fused heterocycles

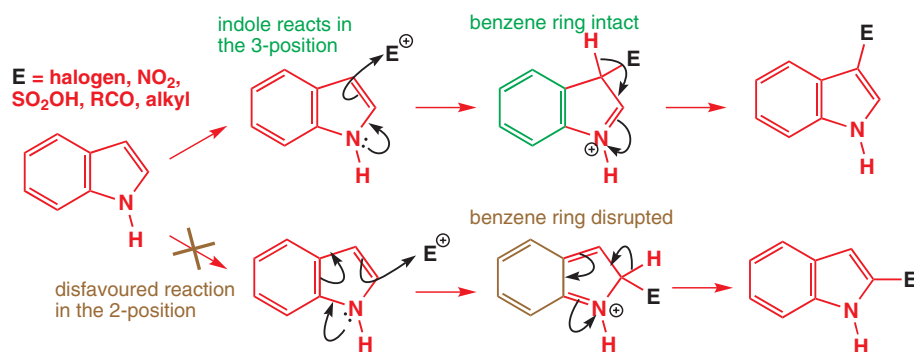
Indoles are benzo-fused pyrroles

Indomethacin and its tetrazole analogue contain pyrrole rings with benzene rings fused to the side. Such bicyclic heterocyclic structures are called **indoles** and are our next topic. Indole itself has a benzene ring and a pyrrole ring sharing one double bond, or, if you prefer to look at it this way, it is an aromatic system with 10 electrons—eight from four double bonds and the lone pair from the nitrogen atom.

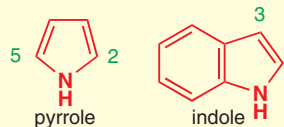


Indole is an important heterocyclic system because it is built into proteins in the form of the amino acid tryptophan (Chapter 42) because it is the basis of important drugs such as indomethacin, and because it provides the skeleton of the **indole alkaloids**—biologically active compounds from plants including strychnine and LSD (alkaloids are discussed in Chapter 42).

In many ways the chemistry of indole is that of a reactive pyrrole ring with a relatively unreactive benzene ring standing on one side—electrophilic substitution almost always occurs on the pyrrole ring, for example. But indole and pyrrole differ in one important respect. In indole, electrophilic substitution is preferred in the 3-position with almost all reagents whereas it occurs in the 2-position with pyrrole. Halogenation, nitration, sulfonation, Friedel–Crafts acylation, and alkylation all occur cleanly at that position.



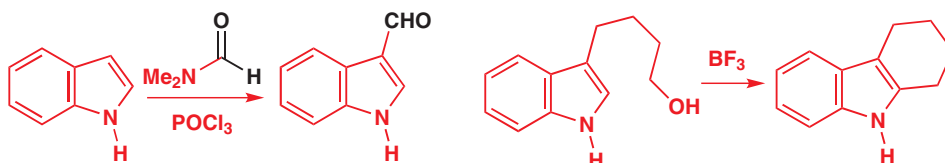
This is, of course, the reverse of what happens with pyrrole. Why should this be? A simple explanation is that reaction at the 3-position simply involves the rather isolated enamine system in the five-membered ring and does not disturb the aromaticity of the benzene ring. The positive charge in the intermediate is, of course, delocalized round the benzene ring, but it gets its main stabilization from the nitrogen atom. It is not possible to get reaction in the 2-position without seriously disturbing the aromaticity of the benzene ring.



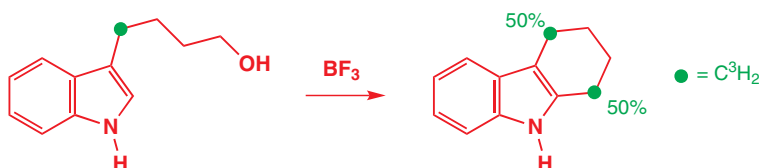
● **Electrophilic substitution on pyrrole and indole**

Pyrrole reacts with electrophiles at all positions but prefers the 2- and 5-positions, while indole much prefers the 3-position.

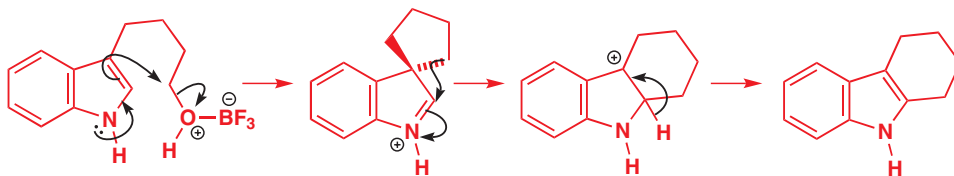
A simple example of electrophilic substitution is the Vilsmeier formylation with DMF and POCl_3 , showing that indole has similar reactivity, if different regioselectivity, to pyrrole. If the 3-position is blocked, reaction occurs at the 2-position and this at first seems to suggest that it is all right after all to take the electrons the 'wrong way' round the five-membered ring. This intramolecular Friedel–Crafts alkylation is an example.



An ingenious experiment showed that this cyclization is not as simple as it seems. If the starting material is labelled with tritium (radioactive ^3H) next to the ring, the product shows exactly 50% of the label where it is expected and 50% where it is not.

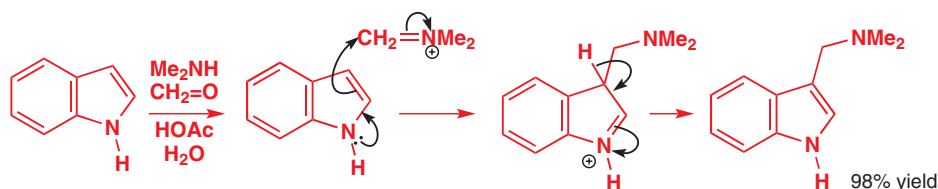


To give this result, the reaction must have a symmetrical intermediate and the obvious candidate arises from attack at the 3-position. The product is formed from the intermediate *spiro* compound, which has the five-membered ring at right angles to the indole ring—each CH_2 group has an exactly equal chance of migrating.

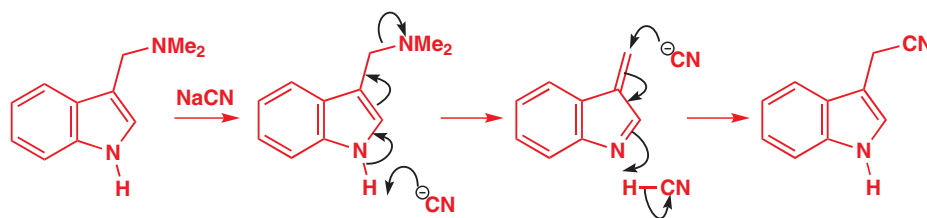


➡ The migration is a pinacol-like rearrangement similar to those in Chapter 36.

It is now thought that most substitutions in the 2-position go by this migration route but that some go by direct attack with disruption of the benzene ring. A good example of indole's 3-position preference is the Mannich reaction, which works as well with indole as it does with pyrrole or furan.



The electron-donating power of the indole and pyrrole nitrogens is never better demonstrated than in the use to which these Mannich bases (the products of the reaction) are put. You may remember that normal Mannich bases can be converted to other compounds by alkylation and elimination (see p. 621). No alkylation is needed here as the indole nitrogen can even expel the Me_2N group when NaCN is around as a base and nucleophile. The reaction is slow and the yield not wonderful but it is amazing that it happens at all. The reaction is even easier with pyrrole derivatives.

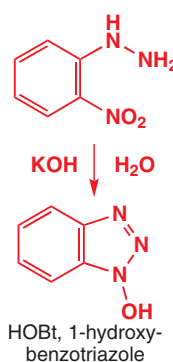
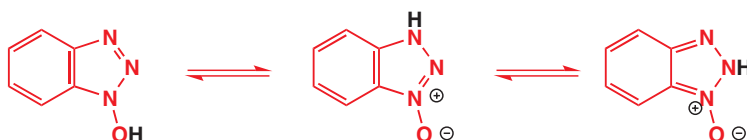


All of the five-membered rings we have looked at have their benzo-derivatives but we will concentrate on just one, 1-hydroxybenzotriazole, both because it is an important compound and because we have said little about simple 1,2,3-triazoles.

HOBt is an important reagent in peptide synthesis

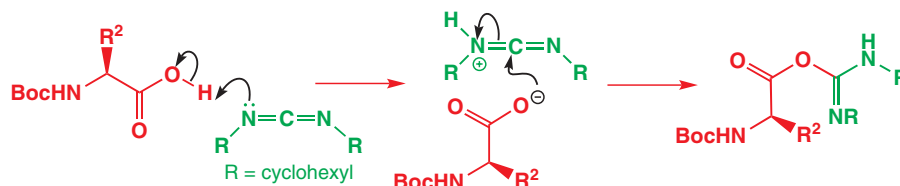
1-Hydroxybenzotriazole (HOBt) is a friend in need in the lives of biochemists. It is added to many reactions where an activated ester of one amino acid is combined with the free amino group of another (see Chapter 23 for some examples). It was first made in the nineteenth century by a remarkably simple reaction.

The structure of HOBt appears quite straightforward, except for the unstable N–O single bond, but we can easily draw some other tautomers in which the proton on oxygen—the only one in the heterocyclic ring—can be placed on some of the nitrogen atoms. These structures are all aromatic, the second and third are nitrones, and the third structure looks less good than the other two.

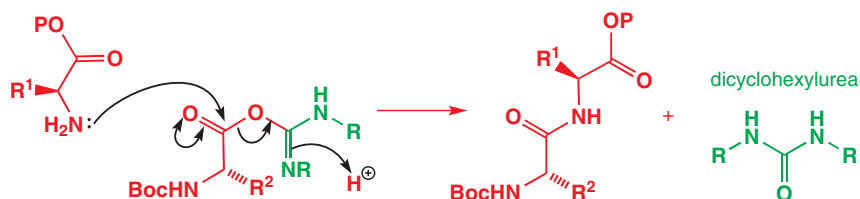


➡ You will meet some nitrone chemistry in Chapter 34.

HOBt comes into play when amino acids are being coupled together in the laboratory. The reaction is an amide formation, but in Chapter 23 we mentioned that amino-acyl chlorides cannot be used to make polypeptides—they are too reactive and they lead to side reactions. Instead, activated amino-esters (with good RO^- leaving groups) are used, such as the phenyl esters of Chapter 23. It is even more common to form the activated ester in the coupling reaction, using a coupling reagent, the most common being DCC, dicyclohexylcarbodiimide. DCC reacts with carboxylic acids like this:

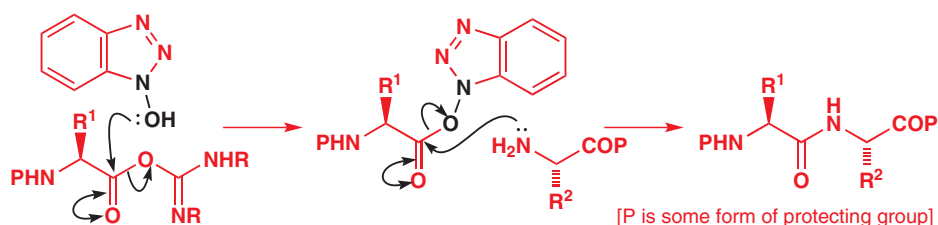


The product ester is activated because substitution with any nucleophile expels this very stable urea as a leaving group.

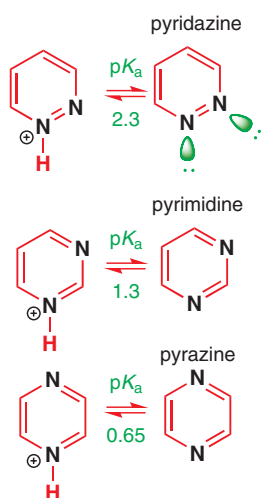


The problem with attacking this ester directly with the amino group of the second amino acid is that some racemization of the active ester is often found. A better method is to have plenty of HOBT around. It intercepts the activated ester first and the new intermediate does not racemize, mostly because the reaction is highly accelerated by the addition of HOBT. The second amino acid, protected on the carboxyl group, attacks the HOBT ester and gives the dipeptide in a very fast reaction without racemization.

➡ You saw in Chapter 26 that the most electrophilic carboxylic acid derivatives are also the most enolizable.

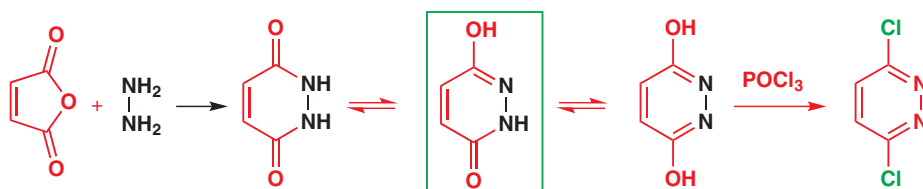


Putting more nitrogen atoms in a six-membered ring

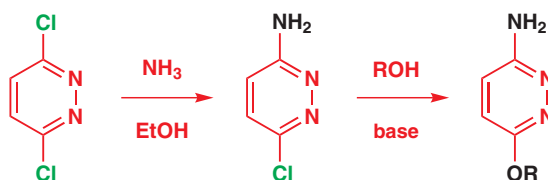


At the beginning of the chapter we mentioned the three six-membered aromatic heterocycles with two nitrogen atoms—pyridazine, pyrimidine, and pyrazine. In these compounds both nitrogen atoms must be of the pyridine sort, with lone pair electrons not delocalized round the ring.

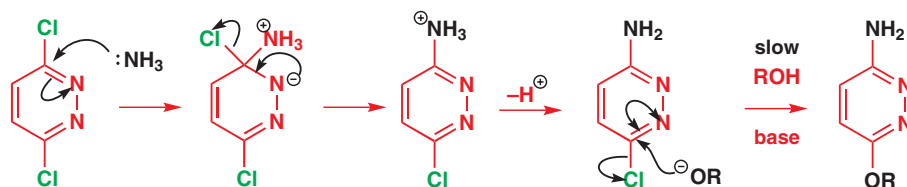
We are going to look at these compounds briefly here. Pyrimidine is more important than either of the others because of its involvement in DNA and RNA—you will find this in Chapter 42. All three compounds are very weak bases—hardly basic at all in fact. Pyridazine is slightly more basic than the other two because the two adjacent lone pairs repel each other and make the molecule more nucleophilic (the α effect again: see p. 513). The chemistry of these very electron-deficient rings mostly concerns nucleophilic attack and displacement of leaving groups such as Cl by nucleophiles such as alcohols and amines. To introduce this subject we need to take one heterocyclic synthesis at this point, although these are properly the subject of the next chapter. The compound maleic hydrazide has been known for some time because it is easily formed when hydrazine is acylated twice by maleic anhydride.



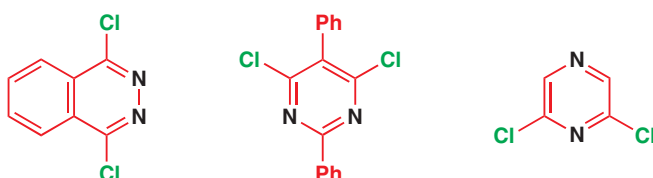
The compound actually prefers to exist as the second tautomer (in the green frame). Reaction with POCl_3 in the way we have seen for pyridine gives the undoubtedly aromatic pyridazine dichloride. Now we come to the point. Each of these chlorides can be displaced in turn with an oxygen or nitrogen nucleophile. Only one chloride is displaced in the first reaction, if that is required, and then the second can be displaced with a different nucleophile.



How is this possible? The mechanism of the reactions is addition to the pyridazine ring followed by loss of the leaving group. When the second nucleophile attacks it is forced to attack a less electrophilic ring. An electron-withdrawing group (Cl) has been replaced by a strongly electron-donating group (NH_2) so the rate-determining step, the addition of the nucleophile, is slower.

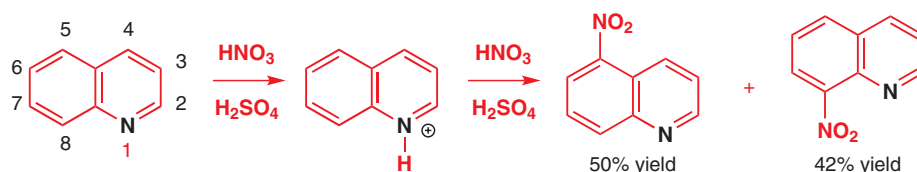
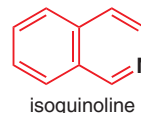
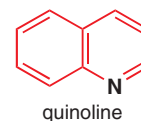


The same principle applies to other easily made symmetrical dichloro derivatives of these rings and their benzo analogues. The nitrogen atoms can be related 1,2, 1,3, or 1,4, as in these examples. The first two are used to link the quinine-derived ligands required for the Sharpless asymmetric dihydroxylation, which will be described in Chapter 41.

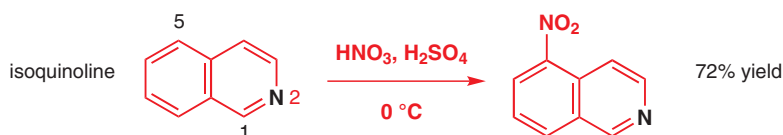


Fusing rings to pyridines: quinolines and isoquinolines

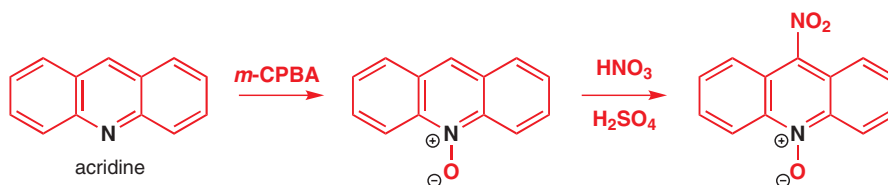
A benzene ring can be fused on to the pyridine ring in two ways, giving the important heterocycles quinoline, with the nitrogen atom next to the benzene ring, and isoquinoline, with the nitrogen atom in the other possible position. Quinoline forms part of quinine (structure at the head of this chapter) and isoquinoline forms the central skeleton of the isoquinoline alkaloids, which we will discuss in Chapter 42. In this chapter we need not say much about quinoline because it behaves rather as you would expect—its chemistry is a mixture of that of benzene and pyridine. Electrophilic substitution favours the benzene ring and nucleophilic substitution favours the pyridine ring. So nitration of quinoline gives two products—the 5-nitroquinolines and the 8-nitroquinolines—in about equal quantities (although you will realize that the reaction really occurs on protonated quinoline).



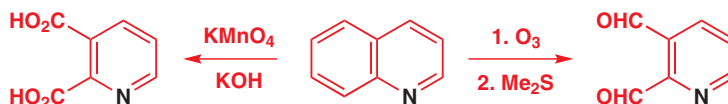
This is obviously rather unsatisfactory but nitration is actually one of the better behaved reactions. Chlorination gives ten products (at least!), of which no fewer than five are chlorinated quinolines of various structures. The nitration of isoquinoline is rather better behaved, giving 72% of one isomer (5-nitroisoquinoline) at 0 °C.



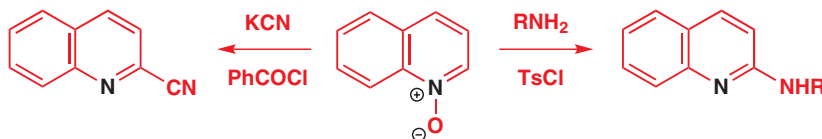
To get reaction on the pyridine ring, the *N*-oxide can be used—as with pyridine itself. A good example is acridine, with two benzene rings, which gives four nitration products, all on the benzene rings. Its *N*-oxide, on the other hand, gives just one product in good yield—nitration takes place at the only remaining position on the pyridine ring.



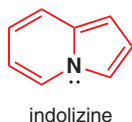
In general, these reactions are not of much use and most substituents are put into quinolines during ring synthesis from simple precursors, as we will explain in the next chapter. There are a couple of quinoline reactions that are unusual and interesting. Vigorous oxidation goes for the more electron-rich ring, the benzene ring, and destroys it leaving pyridine rings with carbonyl groups in the 2- and 3-positions.



A particularly interesting nucleophilic substitution occurs when quinoline *N*-oxide is treated with acylating agents in the presence of nucleophiles. These two examples show that nucleophilic substitution occurs in the 2-position and you may compare these reactions with those of pyridine *N*-oxide. The mechanism is similar.



In considering quinolines and indoles with their fused rings we kept the benzene and heterocyclic rings separate. Yet there is a way in which they can be combined more intimately, and that is to have a nitrogen atom at a ring junction.

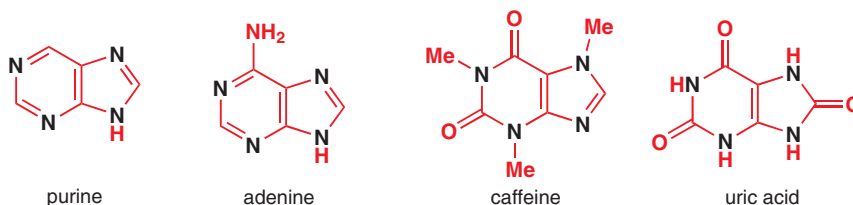


A nitrogen atom can be at a ring junction

It has to be a pyrrole-type nitrogen as it must have three σ bonds, so the lone pair must be in a *p* orbital. This means that one of the rings must be five-membered and the simplest member of this interesting class is called indolizine—it has pyridine and pyrrole rings fused together along a C–N bond. If you examine this structure you will see that there is definitely a pyrrole ring but that the pyridine ring is not all there. Of course, the lone pair and the π electrons are all delocalized but this system, unlike indole and quinoline, is much better regarded as a ten-electron outer ring than as two six-electron rings joined together. Indolizidine reacts with electrophiles on the five-membered rings by substitution reactions as expected.

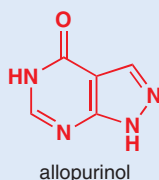
Fused rings with more than one nitrogen

It is easily possible to continue to insert nitrogen atoms into fused ring systems and some important compounds belong to these groups. The purines are part of DNA and RNA, one example is adenine and another is guanine in the box below, but simple purines play an important part in our lives. Coffee and tea owe their stimulant properties to caffeine, a simple trimethyl purine derivative. It has an imidazole ring fused to a pyrimidine ring and is aromatic in spite of the two carbonyl groups.

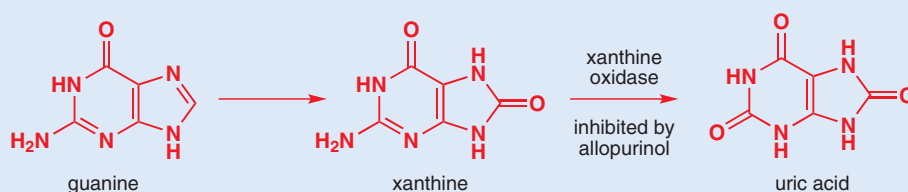


Uric acid, gout, and allopurinol

Another purine, uric acid, occurs widely in nature—it is used by birds, and to some extent by humans, as a way to excrete excess nitrogen—but it causes much distress in humans when crystalline uric acid is deposited in joints. We call the pain 'gout'. The solution is a specific inhibitor of the enzyme producing uric acid and it is no surprise that a compound closely resembling uric acid, allopurinol, is the best.



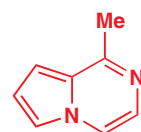
Two of the carbonyl groups have gone and the imidazole ring has been replaced by a pyrazole ring. Purines from DNA are degraded in the body to xanthine, which is oxidized to uric acid. Allopurinol binds to the enzyme xanthine oxidase but inactivates it by not reacting. In fact it imitates not uric acid but the true substrate xanthine in a competitive fashion. This enzyme plays a minor part in human metabolism so inhibiting it is not serious—it just prevents over-production of uric acid.



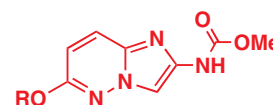
Other fused heterocycles have very attractive flavour and odour properties. Pyrazines, in general, are important in many strong food flavours: a fused pyrazine with a ring junction nitrogen atom is one of the most important components in the smell of roast meat. You can read about the simple pyrazine that provides green peppers with their flavour in the box on the next page.

Finally, the compounds in the margin form a medically important group of molecules, which includes antitumour compounds for humans and anthelmintics (compounds that get rid of parasitic worms) for animals. They are derived from a 6/5 fused aromatic ring system that resembles the ten-electron system of the indolizine ring system but has three nitrogen atoms.

All this multiple heteroatom insertion is possible only with nitrogen and we need to look briefly at what happens when we combine nitrogen with oxygen or in heterocycles.



smell of roast meat



useful medicinal compounds

Aromatic heterocycles can have many nitrogens but only one sulfur or oxygen in any ring

A neutral oxygen or sulfur atom can have only two bonds and so it can never be like the nitrogen atom in pyridine—it can only be like the nitrogen atom in pyrrole. We can put as many pyridine-like nitrogens as we like in an aromatic ring, but never more than one pyrrole-like nitrogen. Similarly, we can put only one oxygen or sulfur atom in an aromatic ring. The simplest examples are oxazoles and thiazoles, and their less stable isomers isoxazoles and isothiazoles.

The instability of the 'iso-' compounds comes from the weak O–N or S–N bond. These bonds can be cleaved by reducing agents, which then usually reduce the remaining functional groups further. The first product from reduction of the N–O bond is an unstable imino-enol. The enol tautomerizes to the ketone and the imine may be reduced further to the amine.

Such heterocycles with even more nitrogen atoms exist but are relatively unimportant and we shall mention just one, the 1,2,5-thiadiazole, because it is part of a drug, timolol.



oxazole



thiazole

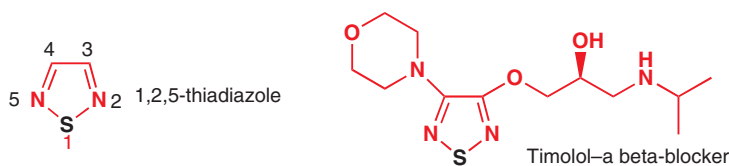


isoxazole



isothiazole

■ Timolol is a β -blocker that blocks one action of adrenaline (epinephrine) and keeps heart disease at bay by counteracting high blood pressure.



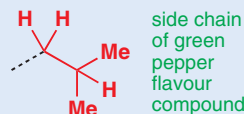
The flavour of green peppers

The discovery of the compound responsible for the flavour of green peppers provides us with a chance to review some spectroscopy. This powerful compound was isolated from the oil of the green pepper (*Capsicum annuum* var. *grossum*). The oil makes up about 0.0001% of the mass of the peppers and the main pepper flavour comes from one compound which is 30% of the oil. It had an even molecular ion at 166 and looks like a compound without nitrogen, perhaps $C_{11}H_{18}O$. But a high-resolution mass spectrum revealed that M^+ was actually 166.1102, which corresponds almost exactly to $C_9H_{14}N_2O$ (166.1106).

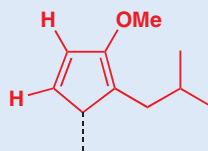
The IR had no OH, NH, or C=O peaks, and the proton NMR looked like this.

δ_{H} , ppm	Integral	Shape	J , Hz	Comments
0.91	6H	d	6.7	Me_2CH^-
1.1–2.4	1H	m	?	
2.61	2H	d	7.0	CH_2CH^-
3.91	3H	s	—	—OMe?
7.80	1H	d	2.4	aromatic
7.93	1H	d	2.4	aromatic

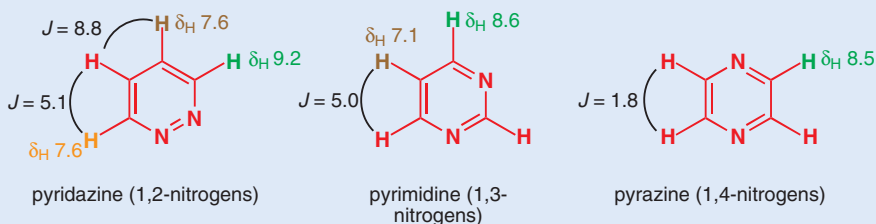
The 'CH' feature in the Me_2CH and CH_2CH signals must be the same CH and it must be the signal at 1.1–2.4 ppm described as a 'multiplet' as it is the only one showing enough coupling. It will be a septuplet of triplets, that is, 21 lines. We can easily reconstruct the aliphatic part of the molecule because it has two methyl groups and a CH_2 group joined to the same CH group.



We also have an OMe group (only oxygen is electronegative enough to take a methyl group to nearly 4 ppm). This adds up to $C_5H_{12}O$. What is left? Only $C_4H_2N_2$ —and no clue yet as to the nitrogen functionality. We also have an aromatic ring that must have nitrogen in it (because there are only five carbon atoms—not enough for a benzene ring!) and the coupling constant between the two aromatic hydrogens is 2.4 Hz. So could we perhaps have a pyrrole ring? Well, no, and for two reasons. If we try and construct such a molecule, we can't fit in the last nitrogen! If we put it on the end of the dotted line, it would have to be an NH_2 group, and there isn't one.

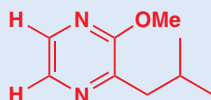


A better reason is that the chemical shifts are all wrong. The protons on an electron-rich pyrrole ring come at around 6–6.5 ppm, upfield from benzene (7.27 ppm). But these protons are at 7.8–8.0 ppm, downfield from benzene. We have a deshielded (electron-poor) ring, not a shielded (electron-rich) ring. From what you now know of heterocyclic chemistry, the ring must be a six-membered one, and we must put both nitrogen atoms in the ring. There are three ways to do this.



The small coupling constant really fits the pyrazine alone and the chemical shifts are about right for that molecule too, although not as far downfield. But we have a MeO group on the ring feeding electrons into the aromatic system and that will increase the shielding slightly and move the protons upfield. This gives us a unique structure.

correct structure
of the main flavour
compound from
green peppers

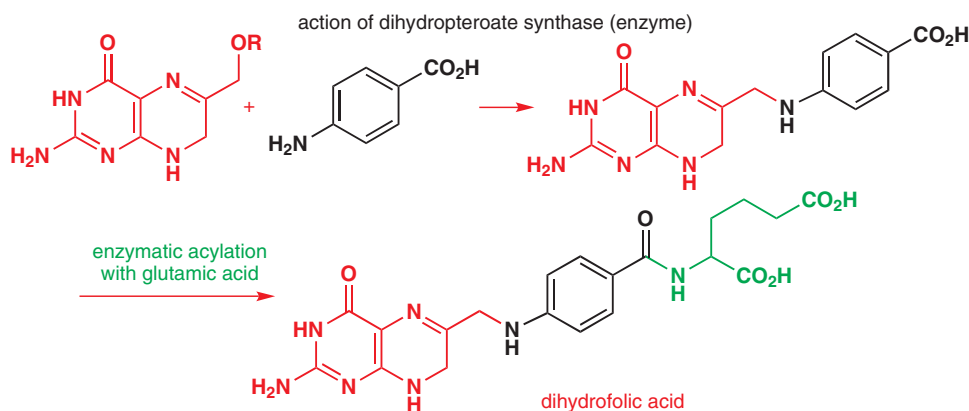


There is only one way to be sure and that is to make this compound and see if it is the same as the natural product in all respects, including biological activity. The investigators did this but then wished that they hadn't! The structure was indeed correct but the biological activity—the smell of green peppers—was so intense that they had to seal up the laboratory where the work was done as no one would work there. Human beings can detect 2 parts in 10^{12} of this compound in water.

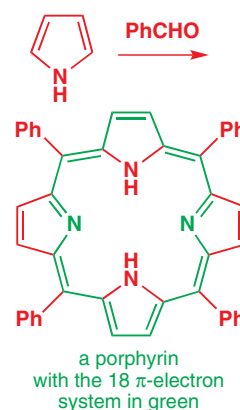
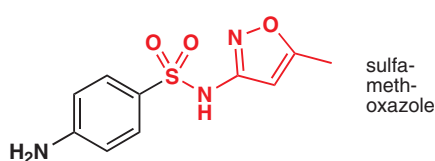
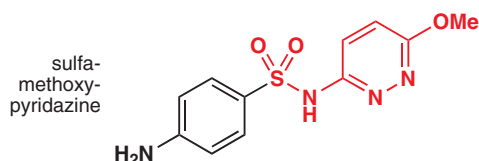
There are thousands more heterocycles out there

But we're not going to discuss them and we hope you're grateful. In fact, it's about time to stop, and we shall leave you with a hint of the complexity that is possible. If pyrrole is combined with benzaldehyde a good yield of a highly coloured crystalline compound is formed: a porphyrin. Now, what about this ring system—is it aromatic? It's certainly highly delocalized and your answer to the question clearly depends on whether you include the nitrogen electrons or not. In fact, if you ignore the pyrrole-like nitrogen atoms but include the pyridine-like nitrogens and weave round the periphery, you have nine double bonds and hence 18 electrons—a $4n + 2$ number. Most people agree that these compounds are aromatic.

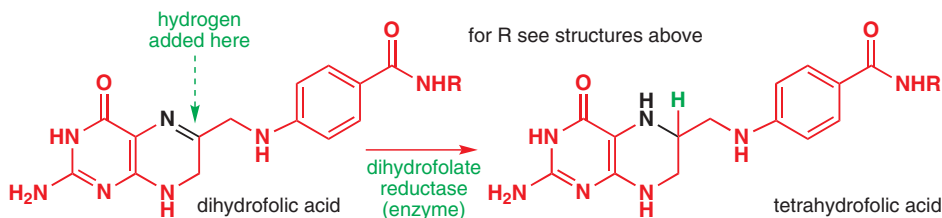
Some heterocycles are simple, some very complex, but we cannot live without them. We shall end this chapter with a wonderful story of heterocyclic chemistry at work. Folic acid is much in the news today as a vitamin that is particularly important for pregnant women, but is involved in the metabolism of all living things. Folic acid is built up in nature from three pieces: a heterocyclic starting material (red), *p*-aminobenzoic acid (black), and the amino acid glutamic acid (green). Here you see the precursor, dihydrofolic acid.



Although folic acid is vital for human health, we don't have the enzymes to make it: it's a vitamin, which means we must take it in our diet or we die. Bacteria, on the other hand, do make folic acid. This is very useful because it means that if we inhibit the enzymes of folic acid synthesis we can kill bacteria but we cannot possibly harm ourselves as we don't have those enzymes. The sulfa drugs, such as sulfamethoxypyridazine or sulfamethoxazole, imitate *p*-aminobenzoic acid and inhibit the enzyme dihydropteroate synthase. Each has a new heterocyclic system added to the sulfonamide part of the drug.

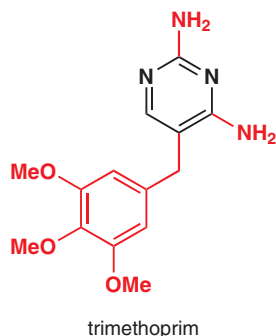


The next step in folic acid synthesis is the reduction of dihydrofolate to tetrahydrofolate. This can be done by both humans and bacteria, and although it looks like a rather trivial reaction (see black portion of molecules), it can only be done by the very important enzyme **dihydrofolate reductase**.



Although both bacteria and humans have this enzyme, the bacterial version is different enough for us to attack it with specific drugs. An example is trimethoprim—yet another heterocyclic compound with a pyrimidine core (black on diagram). These two types of drugs that attack the folic acid metabolism of bacteria are often used together.

We will see in the next chapter how to make these heterocyclic systems and, in Chapter 42, other examples of how important they are in living things.



Which heterocyclic structures should you learn?

This is, of course, nearly a matter of personal choice. Every chemist really must know the names of the simplest heterocycles and we give those below along with a menu of suggestions. First of all, those every chemist must know:



pyridine



pyrrole



thiophene



furan

Now the table gives a suggested list of five ring systems that have important roles in the chemistry of life and in human medicine—many drugs are based on these five structures.

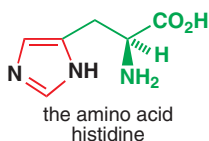
1 Imidazole

the most important five-membered ring with two nitrogen atoms

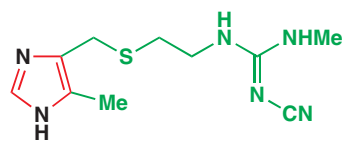


imidazole

part of the amino acid histidine, occurs in proteins and is important in enzyme mechanisms



a substituted imidazole is an essential part of the anti-ulcer drug cimetidine



the anti-ulcer drug cimetidine (Tagamet)
a selective histidine mimic

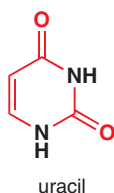
2 Pyrimidine

the most important six-membered ring with two nitrogen atoms

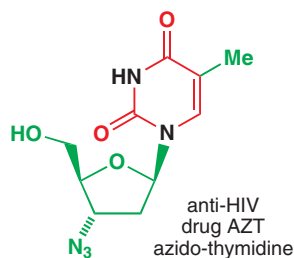


pyrimidine

three functionalized pyrimidines are part of DNA and RNA structure, e.g. uracil

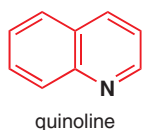


many antiviral drugs, particularly anti-HIV drugs, are modified pieces of DNA and contain pyrimidines

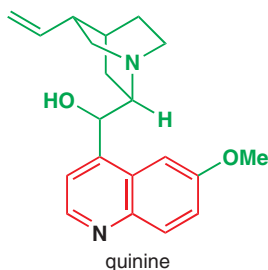


3 Quinoline

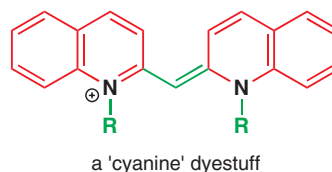
one of two benzo-pyridines with many applications



occurs naturally in the important antimalarial drug quinine

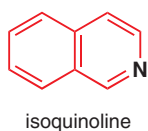


'cyanine' dyestuffs used as sensitizers for particular light wavelengths in colour photography

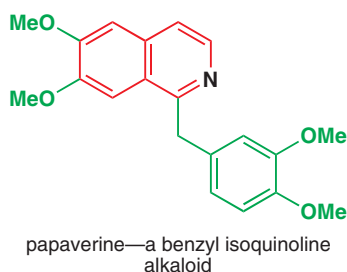


4 Isoquinoline

the other benzo-pyridine with many applications

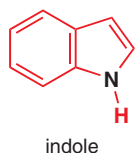


occurs naturally in the benzyl isoquinoline alkaloids like papaverine

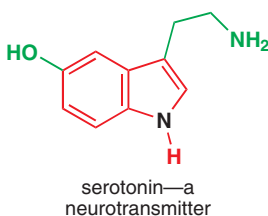


5 Indole

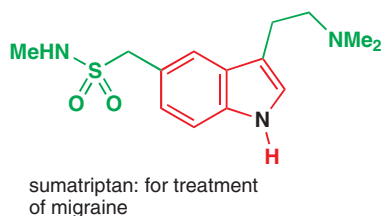
the more important benzo-pyrrole



occurs in proteins as tryptophan and in the brain as the neurotransmitter serotonin (5-hydroxytryptamine)



important modern drugs are based on serotonin, including sumatriptan for migraine and ondansetron, an antiemetic for cancer chemotherapy



Further reading

Basic introduction: *Aromatic Heterocyclic Chemistry*, D. T. Davies, Oxford Primer, OUP, 1992. The best general text on heterocycles is J. A. Joule and K. Mills, *Heterocyclic Chemistry*, 4th edn, Chapman and Hall, London, 2010. S. Warren and P. Wyatt, *Workbook for Organic Synthesis: the Disconnection Approach*, Wiley, Chichester, 2009, chapters 32, 34, and 35.

Reference for the green pepper compound: R. G. Buttery, R.M. Seifert, R. E. Lundin, D. G. Guadagni, and L. C. Ling, *Chemistry and Industry (London)*, 1969, 490.

Diazotetrazole was made by P. B. Shevlin, *J. Am. Chem. Soc.* 1972, 94, 1379 and used in a reaction with buckminsterfullerene by R. M. Strongin and group, *J. Org. Chem.*, 1998, 63, 3522.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>